

The Edge Negotiator: Trustworthy On-Device Small-Language-Model Control of Urban Traffic Signals, with a Provable Accident Audit

WKS2

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Abstract

When a signalised junction is involved in a collision, someone must prove what the controller knew and what it decided. Published language-model controllers are cloud-served and cannot answer that question. This report asks whether a cabinet-sized language model can control real London junctions competently and bind every actuation into a record an investigator can trust, and what fine-tuning contributes. The answer, inside a deterministic shield, is that what fine-tuning reliably moves is *authorship*, not delay.

The full stack runs on one consumer device: a small language model served by Microsoft Foundry Local within 6 GB proposes phases behind a MaxPressure shield and anti-starvation floor, and every decision is Ed25519-signed into a per-junction hash chain, reconstructable after a collision into a cited UK-law evidence pack. A frontier teacher is distilled into 0.6B and 3.8B students with QLoRA, compiled to int4 ONNX, and evaluated offline on a hash-frozen holdout and closed-loop in SUMO on measured London topologies over thirty training-disjoint seeds.

The model's share of logged decision ticks rises from 0.39–0.62 to 0.56–0.77, while its divergence from MaxPressure among the decisions it authors falls from 0.37–0.60 to 0.04–0.17, a deflationary reading I bound. Delay moves only where the network can clear: on a Bloomsbury grid calibrated so that 87% of vehicles complete, the distilled controller beats the fixed-time floor by 8.8% where the stock one loses by 11.9%, an 18.3% head-to-head gain on 30 of 30 seeds ($p=2.0\times 10^{-21}$);

the same calibration on the Euston corridor does not reproduce it. Distillation saturates at 0.6B, a three-seed MaxPressure result is retracted at thirty seeds, seven collision scenarios reconstruct with tampering caught, and a trust ledger catches blatant but not calibrated stealth misreporting.

Author Keywords

Small language models; traffic signal control; on-device inference; knowledge distillation; safety shield; audit trail; Foundry Local.

1 Introduction

1.1 Motivation

Traffic signals are the most consequential actuators most cities own, and the research frontier has begun handing them to large language models: GPT-4-distilled controllers, RL-tuned reasoning models, cooperative agents [14, 36, 33]. Every one is evaluated against a cloud endpoint, yet the junction is an edge environment: no guaranteed backhaul, a hard memory envelope, and a statutory duty, rare among LLM application domains, to account for itself after a collision. This report takes the four research questions (§1.2) together, since an accountability record concerns the certified component only so far as that component authored the behaviour. The work is conducted with and for Microsoft, which fixes the serving stack, Foundry Local on a 6 GB consumer GPU, capping the model ladder at roughly 4B parameters; that constraint is the research object: what *this* deployable envelope can do, not what a datacentre could.

1.2 Research questions

RQ1 Can an on-device SLM (≤ 4 GB, Foundry Local, RTX 2060-class), behind a deterministic safety shield,

match or beat classical baselines (fixed-time, MaxPressure) on measured London topologies in SUMO?

- RQ2** What does distilling a frontier teacher into the SLM buy offline and closed-loop, and at what model scale does the benefit saturate?
- RQ3** Can every actuation be bound into a tamper-evident audit that, after a simulated collision, yields a UK-law-grounded evidence pack rather than an unaccountable log?
- RQ4** Does a trust-coefficient mechanism over neighbour reports withstand honest, blatant, and stealth misreporting?

1.3 Contributions

1. **The authorship finding.** Behind the shield’s envelope, where delay effects are compressed, offline accuracy decouples from closed-loop delay, but fine-tuning raises the SLM’s share of logged decision ticks from 0.39–0.62 to 0.56–0.77 and cuts its divergence from MaxPressure between roughly three- and fourteen-fold; the complement is the anti-starvation floor, which fires on 24 to 43% of ticks with no SLM at all, so what fine-tuning removes is the stock proposer’s excess above that rate. Authorship moves even where delay does not, and I report it with the rival imitation reading stated (§6.6).
2. **A large, seed-unanimous fine-tuning effect and the measurement condition needed to see it.** On a Bloomsbury grid at its default demand, where under 15% of vehicles clear, distillation is worth under two points; calibrated so that 87% clear, the distilled controller beats the fixed-time floor where the stock controller loses to it, an 18.3% head-to-head improvement on 30 of 30 seeds. The same calibration on the corridor reproduces neither the size nor the significance of that effect, so I report a single-network result with its failed replication (§6.5). In the same spirit I retract my own earlier claim: a three-seed sweep showed MaxPressure worse than naive fixed-time on every real London topology, and at thirty paired seeds it is significantly better on Old Street, tied on Bloomsbury and significantly worse on Euston, so its advantage over a naive plan is unreliable on irregular real geometry rather than absent (§6.1).
3. **Scale saturation at 0.6B.** Frontier distillation lifts a 0.6B student to the teacher-agreement ceiling and a 3.8B student lands on statistically indistinguishable numbers offline and closed-loop (§6.3); on the calibrated grid the two scales sit within $\pm 0.38\%$ of each other on delay at 95% confidence, a bounded difference rather than a bare non-rejection, and they reproduce each other’s control trajectory exactly on a tenth of all cells (§6.7). The bottleneck is the distillation ceiling, not capability.
4. **The first documented end-to-end on-device chain** from QLoRA fine-tuning through int4 ONNX compilation to Foundry Local serving of a custom model, with equal aggregate holdout accuracy on the one artifact

and format scored under both CPU and GPU execution providers (§4.2, §6.3). Supporting it: three serving-reliability findings with mitigations (§6.12), and the negative that serving-artifact provenance alone shifts a headline delay result by nine points, which is why every comparison uses an identical-pipeline control (§6.2).

5. **An audit-to-law pipeline verified end-to-end on seven scenarios.** Seven collision scenarios reconstructed from signed, hash-chained records, with tampering caught and the governing UK rule inferred deterministically from verified facts into cited evidence packs rather than verdicts (§6.8). Supporting it: a scoped, discount-only trust mechanism whose ledger catches most blatant misreporting and is evaded by a calibrated stealth attacker, at three seeds and with gating off (§6.9); and the negative that explicit in-prompt policy rescues a stock model’s recall but not its safety in an authorisation re-test, with the fine-tuned students format-captured on that adjacent task (§6.11).

1.4 Scope and thesis statement

The locked thesis question is: *is an on-device Foundry-Local SLM traffic controller that matches or beats classical control on real London topology, while producing a provable accident audit, possible at all, and at what model scale and configuration?* This is a feasibility and scale-threshold study: everything runs in SUMO on three real London topologies plus synthetic grids, only Euston demand magnitude is measured, realisations are synthetic throughout, and the comparators are classical rather than trained deep-RL, since the target is feasibility rather than a new state of the art in delay.

In answering it I argue something narrower than feasibility: inside a deterministic shield, what fine-tuning reliably moves is *authorship*, not delay. Stock students already author 39 to 62 per cent of logged decision ticks, a majority on four of the five cells; the fine-tuned students author more on every cell I tested, though the size of that shift varies widely and I did not isolate its mechanism. Delay moved materially only on the one network I calibrated to clear, and not on every network that clears: the corridor I calibrated did not follow. That gap is the thesis: an accident audit concerns the certified model only so far as the model authored the decisions, so authorship is the property that has to hold. §6 tests both halves against both classical comparators; §7 reports where each holds, and §8 where it fails.

1.5 Report structure

§2 positions the work; §3–5 give architecture, engineering and protocol; §6 reports the evidence in the order it was built; §7–8 answer the research questions.

2 Background and related work

2.1 Classical signal control and its guarantees

Deployed urban signal control rests on two families. Fixed-time control serves a cycle engineered offline from historical counts. It optimises nothing online, but it is deterministic, starvation-free and indifferent to sensing noise. Its adaptive rival MaxPressure [27], the canonical strong baseline, serves at each intersection the phase of greatest “pressure” (upstream minus downstream queues, weighted by turn ratios), a decentralised rule Varaiya proved throughput-optimal for every stabilisable demand. That guarantee rests on four assumptions, and read against the backpressure literature (Appendix B.7) each fails structurally in central London. The proof assumes unbounded queue storage; short links violate it, and the finite-capacity repair [7] is not part of standard MaxPressure. The only stability result for a deployable controller holds under heavy load [6], which measured diurnal demand sits below for most of the day. Every favourable empirical figure comes from a grid. And service in the theorems is binary and phase-determined, which give-way, gap-acceptance and gyratory junctions break. I do not measure the first two conditions on my networks.

The field’s own tables corroborate this. AttendLight [20] reports MaxPressure *losing* to naive fixed-time by 4.6–18.6% on some T-junctions and two to four times on its two four-way layouts with unequal lane counts (INT10, INT11), a geometry-specific collapse. Against this, at least six independent studies [29, 14, 36, 32, 33, 8] show it beating fixed-time by 7–58% on grid-like CityFlow and SUMO networks, including nominally “real” Manhattan, Jinan and Hangzhou topologies, all on synthetic or replayed-uniform demand. My measurements (§6.1) fall between: the sign changes with the geometry, while my synthetic grids behave conventionally ($n=3$). That pre-empts the objection that my baseline is broken, and it stops short of the stronger reversal an earlier three-seed sweep had suggested. From a targeted rather than systematic review, I find no prior work testing MaxPressure against fixed-time on measured UK geometry under measured demand.

2.2 Learning-based signal control

Deep reinforcement learning dominates the literature classical control left behind: phase-competition invariances in FRAP [35], graph-attention coordination in CoLight [29], cross-layout transfer in AttendLight [20], all surveyed by Wei et al. [28]. The results are strong on the benchmarks the community chose, almost uniformly CityFlow grids with synthetic or replayed demand, typically single-run, so headline numbers carry no variance estimates. Two further limits motivated the turn to language models: trained policies are opaque, giving no account of *why* a phase was served, and the sim-to-real gap goes unexamined in an evaluation monoculture of shared simulators and demand models. This

report does not compete as an RL method: it inherits the framing (phase selection on a fixed interval) and rejects the evaluation habits.

2.3 LLMs and SLMs for traffic signal control

LLMLight/LightGPT [14], the closest prior work, distils GPT-4 rationales into LoRA-tuned students down to 0.5B, yet serves them on four A800-80GB GPUs and evaluates single-run on CityFlow, without seeds or confidence intervals. Traffic-R1 [36] trains a 3B model with GRPO, with a ten-intersection production deployment and evidence that SFT degrades the base model’s general benchmarks; its own future work concedes edge deployment remains unrealised, the artefact this report builds. DGLight [32] (a dissertation preprint, not peer-reviewed) tunes an 8B model on dual H100s and calibrates the field: the LLM-versus-MaxPressure travel-time gap it and its peers report spans 1.1–4.2%, narrower than the seed variance I measure on the corridor topologies and that no competitor tests for. CoLLM-Light [33] scales to 196 intersections on server GPUs and finds its fine-tuned 8B model beating a stock 671B model, the single data point for distillation over scale that motivated my central bet. CuraLight [8] is the one fine-tuning work on SUMO, but its 12B model needs roughly 44GB of VRAM against my 6GB budget. iLLM-TSC [21] inverts my safety topology on a single synthetic four-way intersection, an RL agent proposing and a cloud GPT-4 vetoing, with a retry loop falling back *silently* to the unvetted action; its degraded-communications threat model is a scenario axis I do not test. LATS [34] deploys nothing generative, distilling an LLM teacher into MLP/GRU students, and its pure-LLM baselines perform worst, corroborating my finding that stock small models are weak deciders. Across all seven, every system serves from cloud or server GPUs, every headline table is single-run or at best $n \leq 10$ without statistical tests, none pairs control with an audit mechanism, and none evaluates on measured UK topologies.

2.4 Small language models, distillation, and quantisation

I adopt the SLM survey [19] and on-device review [30] framing: this system is *edge-only* (no cloud fallback), reported against the review’s indicator template (time-to-first-token, VRAM, latency percentiles); neither survey contains a traffic or control-loop case study. The nearest on-device comparator, Octopus [2], is a 2B function-calling model at 1.1–1.7s on Android, comparable to the 0.94–1.75s per-cell median decision latency of my deployed students (Table 10). The surveys corroborate a reliability floor: Qwen2.5’s report [23] shows instruction-following (IFEval) collapsing from 84.1 at 72B to 27.9 at 0.5B, consistent with the 45–69% decision accuracy of my stock 0.6B model, while the Qwen3 report [24] complicates this, its 0.6B posting respectable IFEval scores yet proving the weakest decider of the Qwen3 family I

served, parsing 93–100% strict JSON while choosing wrong phases; generic instruction-following benchmarks do not measure numeric decision quality.

On distillation there is a tension: Orca [18] argues small models need rich explanation traces, whereas my distillation is label-only. Three defences: the task is a bounded, closed decision schema closer to classification than open generation, for which the knowledge-distillation survey [31] endorses labelling-plus-SFT; Orca itself concedes small models excel in constrained settings; and my student nears the teacher ceiling (97.7% against 97.3–99.3% inter-model label agreement on 600 cross-checked items). I concede what Orca implies: out-of-distribution generalisation and natively generated audit rationales are untested risks. The jump from 55% to 97.7% has precedent [10, 26]; LoRA [11] and QLoRA [4] justify rank 16, though their evidence stops above 125M parameters, and I flag the extrapolation.

Quantisation is the larger tension. QLoRA trains adapters on a 4-bit base and serves unmerged, whereas I train QLoRA-style, merge to bf16, then apply post-hoc int4 round-to-nearest: two quantisation steps, and QLoRA’s authority does not transfer to the second. The literature predicts trouble: GPTQ [5] shows small models are the hard case for round-to-nearest, AWQ [17] finds the gap narrowing at 4-bit with grouping, and post-hoc quantisation after fine-tuning has been measured losing 9.4% relative even with GPTQ [25]. My contrary finding is therefore scoped narrowly: equal aggregate held-out decision accuracy, not output identity, under weight-only int4, on a near-converged narrow-schema checkpoint, with a discrete accuracy metric coarser than perplexity. Forgetting-scaling results [13] predict general-capability drift at my very low training loss, carried as a limitation (§7.3) that Traffic-R1’s fine-tuning-paradigm result independently motivates. No published work supplies per-decision reliability data for a generative model at 0.6B, int4-quantised, inside a closed control loop.

2.5 Safe-by-construction shields

The canonical formalism is shielding [1]: a reactive system synthesised from an LTL safety specification and a finite MDP abstraction by solving a two-player safety game, placed preemptively or post-posed, with proofs my system does not carry: correctness for all abstraction-consistent environments, minimal interference, and preserved learner convergence; the authors also warn the shield must remain active after learning, which mine does. Two weaker rungs sit below it: invalid-action masking [12], which proves the masked update is a valid policy gradient, scales where penalties fail, and leaves agents trained masked largely valid unmasked; and neurosymbolic masking [9], which learns the symbolic grounding when constraints are unknown and points back towards temporal-logic specifications. My shield is *not* a shield in the Alshiekh sense: it composes three weaker but sound pieces, enforced by code rather than

proved from a specification (§3.3), so there is no minimal-interference proof and no lookahead, my floor reacting *at* the starvation threshold where a synthesised shield acts before violation becomes unavoidable; the upgrade path is cheap at single-junction scale and is named as future work. My topology inverts iLLM-TSC’s (§2.3, §3.1), which a 6GB edge budget makes viable; I did not implement the inverse to measure the difference. No prior LLM-based signal-control work states the formal status of its safety layer at all.

2.6 Audit, transparency, and legal grounding

My headline thread, that an edge controller should produce a provable account of its decisions, sits in a three-way gap in the trust literature. On the attack side, Qu et al. [22] show roughly 6% of vehicles feeding falsified data to an undefended learning controller: the colluders cut their own waiting time by 92.5% and raise everyone else’s by 62.7%, and their most dangerous result, that a *calibrated* lie beats a maximal one, motivates my stealth-liar scenario; they name real-time anomaly detection as unbuilt future work. On the integrity side, the blockchain architecture of [16] rejects 100% of spoofed inputs at ~39ms but never names a signature scheme and concedes failure when validators collude. Most fundamentally, input integrity is not decision accountability: it certifies what the controller *heard*, not what it *decided* or why.

My audit design therefore borrows from outside traffic. Certificate Transparency [15] makes tampering detectable without trusting the log operator, through an append-only Merkle log with inclusion and consistency proofs, at a per-decision cost, a signature and a hash, that fits the edge budget where a validator network does not; my chain inherits the property only once its head is externally witnessed (§3.4). The legal layer is modest: Dahl et al. [3] profile pervasive hallucination when language models answer legal questions, ruling out letting any model assert fault, least of all a 0.6B one, so my system emits a *cited evidence pack* (§3.4) against a UK-law knowledge base in which authority-side liability is legally weak and arises essentially only from conflicting-green misfeasance, driver-fault doctrines are strong, and emergency-vehicle exemptions are conditional. I am aware of no prior signal-control system that signs and chains individual control decisions, or that mounts a defence-side trust mechanism, though I claim this from a targeted rather than systematic review.

3 System design

A learned controller at a junction must be *optional for safety and mandatory for accountability*: the deterministic control path must be complete without the model, and every actuation must be provable afterwards, whoever authored it. This section presents the architecture realising that premise and defines the three structural quantities (authored share,

proposal acceptance, divergence rate) that carry the report’s central finding.

3.1 Architecture overview

Each junction runs an identical control loop (Figure 1) on a fixed decision interval (≈ 10 s simulated). Per tick, local state (per-phase queue and waiting measurements from the junction’s own sensors) is rendered into a prompt; the on-device SLM served by Foundry Local proposes a phase, which a deterministic parser extracts and the safety shield (§3.3) masks, validates and possibly substitutes. A congestion gate serves MaxPressure whenever total halting demand falls below a threshold (two vehicles by default), reserving the learned component for the states where it can matter; I ran no gate-disabled arm at evaluation power. The served phase actuates the lights, and a signed audit record joins the run’s hash chain under the junction’s key (§3.4). MaxPressure is computed every tick regardless and is at once the shield’s reference, its fallback and the strong baseline, so the learned component costs one model call and its marginal *safety* risk is bounded by construction; its delay risk is not, since a stock proposer behind the same shield loses 22.7% to MaxPressure on the calibrated grid. Three consequences follow. Graceful degradation is the default: a model timeout, garbage output or a dead runtime (all observed, §6.12) leaves the junction running MaxPressure with the failure recorded. The architecture inverts iLLM-TSC [21] (§2.5), the LLM proposing and a local $O(\text{phases})$ verifier vetoing, since that path must be cheap, local and always available (I did not implement the inverse). And with the deterministic path complete, the SLM’s value is measurable as a *difference* against the same loop with the proposer disabled.

3.2 The SLM decision agent

The agent contract is minimal: given a system instruction and a rendered state, the model must return exactly `{"phase": N}`, at temperature zero over an OpenAI-compatible endpoint, with parsing falling through three tiers (braced JSON, keyed, bare integer); the offline evaluator adds a strict-JSON tier and records the tier per response, so format discipline is measured, not assumed. Four prompt formats act as graded capability probes: **myopic** exposes only per-phase halting counts, answerable by a counting heuristic; **sota** adds delay-aware context (accumulated waiting), the state a delay-optimal policy needs; **prediction** adds short-horizon arrival forecasts; **coordination** adds neighbour notes. The same rendering code serves offline evaluation and closed-loop control, and whether offline accuracy predicts loop outcome is itself a result (§6.4).

3.3 The safety shield

The shield composes three mechanisms, named with the formal literature’s vocabulary (§2.5). **Preemptive action masking** [12]: the model chooses from a fixed menu of

conflict-free phases, so no proposal can create a conflicting green. **Post-posed default substitution**: an invalid, un-parseable or absent proposal is replaced by the MaxPressure phase and recorded as a served decision the SLM did not author. **Anti-starvation runtime monitor**: a minimum-service floor force-serves any green phase passed over for more than a bounded number of consecutive intervals, overriding both the SLM and MaxPressure, except while an admissible emergency preemption holds the signal. The floor is best-effort rather than proved. Only one phase can be served per interval, so when two reach the threshold together the code records the residue as a violation. This is a sound heuristic composite, not a correct-by-construction shield in the Alshiekh sense [1]: there is no synthesis from a temporal-logic specification and no minimal-interference proof, a limitation with an upgrade path (§7.4). It still enforces, for every arm including the baselines, no conflicting greens and no approach starved beyond the floor except where the code records a violation.

Three measured quantities are defined on this structure, and their denominators matter. Every logged decision tick is labelled by *who authored the phase actually served*: the SLM, a MaxPressure substitution, or the anti-starvation floor.

Authored share is $\text{slm_served} / \text{decisions}$, the quantity the audit argument needs: the fraction of *logged decision ticks* whose phase originated with the SLM. The denominator is the ticks the controller logged. Those are the ticks where the congestion gate admitted the SLM, plus gate-skipped ticks on which the anti-starvation floor changed the served phase. Intervals the gate skipped without a phase change are served by MaxPressure and counted on neither side. The ratio therefore understates neither arm, but it is a share of *decisions taken*, not of wall-clock signal time.

Proposal acceptance is $\text{slm_served} / \text{slm_valid_proposals}$: of the ticks where the SLM returned a well-formed proposal, the fraction served rather than displaced. In practice the anti-starvation floor is what displaces it.

Divergence rate is $\text{diverged_and_served} / \text{slm_served}$: of the decisions the SLM authored, the fraction whose phase differed from what MaxPressure would have chosen. It measures how far the served policy departs from the classical reference, *not* how often the shield vetoed the model. Masking makes such a veto structurally impossible for unsafe actions, and substitution fires on a negligible fraction of ticks in the fine-tuned and stock 0.6B arms (§6.6).

None of the three counts unsafe actions blocked, since the substitution path catches only malformed or absent output. Reporting all three separately prevents the error of reading a high acceptance rate as a high share of authorship.

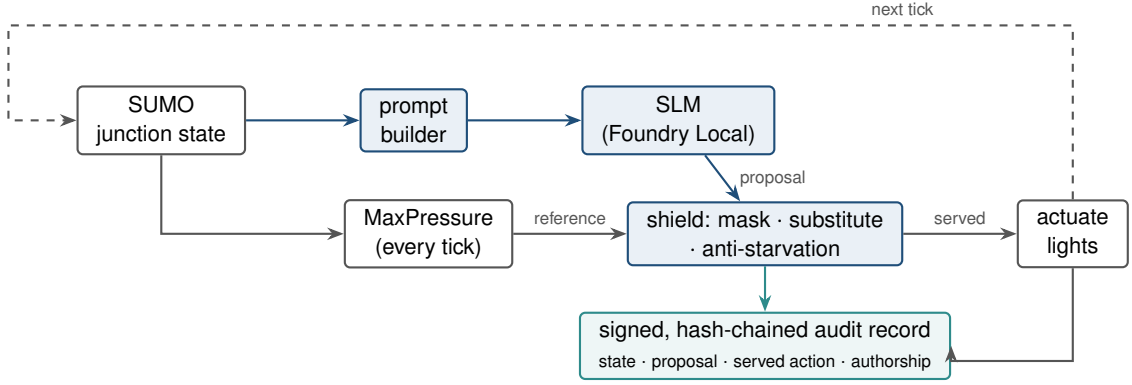


Figure 1: The junction control loop: the deterministic path (grey) is complete without the SLM, the learned proposer (blue) attaches to it, and every served decision lands in the signed audit chain (teal), whoever authored it.

3.4 Audit trail and crash reconstruction

The audit layer is designed backwards from what an investigator asks after a collision: *what did the controller know, what did it decide, who authored it, and can the record be trusted?* Every logged decision tick appends a record containing the halting-queue state vector, the model’s parsed proposal, the shield’s proposal, the served action, authorship (SLM / shield-substitution / anti-starvation) and the override flag; model identity and per-call latency are recorded once at run level. Records are Ed25519-signed per junction and hash-chained into a single append-only chain per run; chain and signature verification run end-to-end after every experiment, and the threat model includes an imposter re-signing attack (caught in all evaluated scenarios, §6.8). The chain head can be anchored to external witnesses in a Certificate-Transparency-style quorum [15], with a permissioned-ledger anchor implemented as one optional witness rather than as a contribution.

On top of verified records sits the reconstruction layer: a UK-traffic-law knowledge base (statutes, case law, the conditional emergency-vehicle exemption; Appendix C) against which a collision scenario’s verified facts are matched deterministically. The output is a *cited evidence pack*, never a verdict: the governing rules, the verified facts triggering them, and the fault anchor each rule implies, which answers the legal-hallucination risk [3] of §2.6 by keeping the legal knowledge a curated, citable database and leaving the adjudicative act to humans. My scenarios use synthetic actors on real topology, so no real data subject exists; the residual legal exposure of a real deployment is in §7.3. The audit’s value depends in turn on authorship (§3.3): an evidence pack over a deployment whose decisions are mostly floor overrides attributes the behaviour to a fairness timer, not to the learned component, so the audit is about the model in proportion to what the model authored, and fine-tuning raises that share (§6.6).

3.5 Trust-coefficient mechanism

Coordination requires acting on neighbour reports, which a compromised neighbour can abuse. The trust mechanism prices that risk with three commitments. **Local sensing is the ultimate truth:** a junction’s own sensors are never doubted, and a neighbour claim is scored against what local sensing later verifies, within a claim lifetime after which it is provisionally discounted. **Asymmetric update:** verified truths earn trust slowly (prior 0.5, ten consecutive truths to reach 0.9) while a detected lie from a high-trust source collapses it in one step (lie factor 0.25), because in infrastructure trusting a liar costs more than doubting an honest peer. **Discount-only wiring:** trust can strip a caught liar’s power to corroborate a preemption but never *add* an action on zero local evidence, so discounting alone cannot make safety worse, and a real ambulance within sensing range is preempted on the junction’s own detection. A truthful report verified late is vindicated retroactively. The adversarial evaluation is in §6.9.

3.6 The fine-tuning axis

The final axis asks what changes when the proposer is fine-tuned rather than stock. The pipeline is label distillation. A frontier teacher labels junction states sampled from training-reserved seeds, and a second frontier model re-labels four audit samples of the teacher’s output (600 states, 98.5% agreement, §4.3). A deterministic sanity verifier screens the teacher’s own answer rather than the state, rejecting a label that serves an empty phase while other approaches are waiting or names a phase outside the junction’s range. The teacher is therefore second-guessed on exactly two degenerate cases and never by a weaker rule competing on the decision itself. The four prompt formats are rendered from three separately sampled and labelled state pools into one mixed training set; training is QLoRA [4], and the merged model is compiled to weight-only int4 ONNX and served through the same runtime as every stock model (§4.3, §4.4). Three rules protect the comparisons (§5.3, §5.4): a holdout hash-frozen before any training, closed-loop evaluation seeds



Figure 2: The Bloomsbury network in SUMO at $t=420$ s under the calibrated 40% demand (seed 1): nine signalised junctions extracted from OpenStreetMap, 112 vehicles in motion.

disjoint from training seeds, and a control arm of the same base model compiled through the identical int4 pipeline, so the only manipulated variable on the 0.6B axis is the weights; the 3.8B student has no identically-compiled stock control. Ablation students, leave-one-topology-out and four format specialists trained under the same recipe, test memorisation and specialisation hypotheses.

4 Implementation

The engineering that made the design measurable: the simulated London networks, the on-device serving stack, the distillation and compile chain, and the experiment harness. Appendix A tabulates the reproducibility pins.

4.1 Simulation environment and topologies

All control runs in SUMO with TraCI. Six topologies span a regular-synthetic to irregular-real axis along which §6 finds the largest differences, without isolating which property carries them: two synthetic grids (3×3 , 4×4), a Bloomsbury grid from OpenStreetMap, the Euston A501 corridor and its peak-hour variant, and the Old Street complex junction. Euston’s AADF-derived demand carries the A501’s measured DfT vehicle mix but an assumed hourly profile, the peak-hour variant rescaled to the measured DfT peak-hour magnitude. Old Street’s nine signals include a single five-arm, seventeen-movement junction, demand-calibrated after default random demand gridlocked it, as were the Bloomsbury grid and Euston peak-hour corridor (§6.5). Only Euston’s demand magnitude is measured, the other two real maps being randomTrips draws (§7.3). Figure 2 shows the Bloomsbury network under the calibrated demand.

4.2 On-device serving with Foundry Local

Serving is Microsoft Foundry Local (pinned 0.8.119) on an RTX 2060 with 6 GB VRAM, consumer hardware of roughly signal-cabinet class and a hard project constraint: it caps the ladder at ~4B (7B-class models OOM the runtime,

recorded), which makes the scale-saturation result (§6.3) deployment-relevant rather than academic.

Custom models serve through two execution providers. The CPU path works out of the box; WebGPU requires editing the compiled model’s `genai_config.json` provider options and delivers about $1.6 \times$ lower latency at *equal aggregate holdout accuracy* on the one artifact and format where both providers were scored (§6.3), which licensed running every sweep GPU-served. That comparison is within one artifact, not a claim of absolute speed: my pipeline-compiled 0.6B artifacts serve at roughly 2.6 times the median latency of the vendor catalogue build of the same weights (§6.12), a real cost of controlling the compile chain. Windows ML CUDA autoregistration requires a newer Windows build than this machine’s, making WebGPU the effective GPU path. Three reliability findings from sustained unattended serving are mitigated in harness (§4.5) and reported in §6.12.

4.3 Distillation dataset and training

From eight training-reserved seeds ($\{3, 7, 11, 29, 42, 57, 91, 123\}$) I sampled 48,137 unique junction states across three separately sampled pools: 16,988 delay-aware states that also render the privileged myopic rows, 18,596 prediction and 12,553 coordination. A frontier teacher labelled them, and a second frontier model re-labelled 600 of them in four audit samples of 150, agreeing on 98.5% (97.3–99.3% per sample). Rendering the four prompt formats over those pools yields 59,533 training examples and a hash-frozen holdout of 5,580 (HOLDOUT_MOD=12), rendered family-aware (ChatML with an empty think block and no-think marker for Qwen3, Qwen2.5’s ChatML without the marker, Phi’s `<|system|>...<|end|>` convention), with loss on the completion only, targeting exactly `{"phase": N}`. Training is QLoRA $r=16$, NF4 double quantisation, fp16 compute. The 0.6B student trained on the serving machine itself (13.5 h on the RTX 2060), so the full loop from teacher to served controller fits on one consumer device; the 3.8B student and its five ablation variants trained on a single UCL RTX 4090 (2.3 h each, 8.9M trainable parameters, 0.23%).

4.4 Compile-and-serve chain

Six steps take merged bf16 weights to a served endpoint: QLoRA merge $\rightarrow$ int4 RTN compile $\rightarrow$ config patch $\rightarrow$ manifest synthesis $\rightarrow$ cache registration $\rightarrow$ WebGPU provider swap. That chain is, to my knowledge, the first documented end-to-end recipe for serving a custom fine-tuned model through Foundry Local; Appendix A records each step with the failure that established it.

4.5 Experiment harness

Two harnesses produced the closed-loop sweep and offline-evaluation numbers: the **factorial runner** executes model $\times$ format $\times$ topology $\times$ seed cells with per-cell raw JSON

artifacts, a pre-flight serving probe per cell and multi-pass retry for deferred cells, and the **offline evaluator** scores holdout rows against the served endpoint; standalone probes supplied the fixed-time floor, emergency preemption, the attack battery, the crash-law audit, the authorisation re-test and the retention probe. Both are fail-loud, resumable and recoverable in the senses Appendix A defines, and survived a six-week unattended campaign including two service restarts, one machine reboot and one mid-write power loss (one corrupted cell, detected by JSON validation and re-run).

5 Experimental methodology

What I measured, how each comparison was controlled, and the statistical decision rules committed to before the experiments ran. The stance is fail-loud: harness errors abort or are counted, never silently scored, and null results are reported as findings.

5.1 Evaluation axes and metrics

I evaluate along four axes. **Closed-loop control quality.** The primary metric is mean network delay: the mean, over all vehicles that *departed* into the network, of total time in network (SUMO’s tripinfo duration) over a fixed 1,200-step horizon, a completed trip contributing its full travel time and a vehicle still running at the horizon the time it has accrued. Every controller cell is paired with a fixed-time baseline on the *same topology and seed*, so the reported quantity is the paired relative delay change; completed-vehicle throughput, teleport count and undeclared backlog guard against delay “improved” by starving demand. **Offline decision quality.** Accuracy against frozen frontier-teacher labels on a held-out set of junction states at int4 precision exactly as served, plus strict-JSON rate (responses matching {"phase": N} exactly) and parse rate (any recoverable phase). **Structural authorship.** Authored share, proposal acceptance and divergence rate (§3.3), reported alongside the decision composition itself. **Serving viability.** Decision latency (p50/p99) against the 10 s decision interval, and a log of serving-reliability events with their mitigations.

5.2 Statistical protocol

Every powered closed-loop cell is replicated over $n=30$ seeds per topology; the exploratory sweep that preceded them ran three seeds per cell and is labelled as such throughout. The test for a delay effect is a one-sample t -test on the per-seed relative change against each baseline, computed on the same quantity as the reported CI and pooled across topologies ($n=90$) where the hypothesis is topology-independent. The explanatory ANOVA uses the project’s pure-mathematics regularised incomplete beta, so no library version can silently change it; the closed-loop t and Wilcoxon p -values are SciPy’s, recomputable from the committed per-seed JSON;

effect sizes (partial η^2) accompany F statistics, and headline comparisons state direction, magnitude and p together. Two commitments were made in advance: the honest-null policy, under which a non-significant result at $n=30$ is reported with its magnitude and uncertainty rather than suppressed or re-run until significant; and no mid-experiment stopping, cell counts being fixed by the factorial design before launch, with the resumable runner (§4.5) re-running only cells that produced no measurement, never cells whose answer was unwelcome.

5.3 Seed-disjointness and holdout freezing

Two leakage paths exist between training and evaluation; the first is closed by construction, the second narrowed, not closed. **State-level leakage.** The distillation dataset is built from junction states sampled on eight reserved seeds (Appendix A); the offline holdout is split from those pools by a deterministic hash of the state content, frozen before any training example was rendered, so no training example, under any format rendering, shares a state with a holdout example. **Trajectory-level leakage.** Closed-loop evaluation uses thirty seeds fully disjoint from the training seeds, $\{1..30\} \setminus \{3, 7, 11, 29\} \cup \{31, 32, 33, 34\}$, since a controller fine-tuned on states from seed s has seen a compressed replay of that seed’s trajectory. The route files are static, with fixed vehicle identities and departure times, and the SUMO seed varies microscopic car-following and lane-change randomness rather than the demand itself, so disjointness narrows this path without closing it: a held-out seed is a different realisation of the same demand, not a different demand. Head-to-head arm comparisons are unaffected, since both arms run the identical route file and seeds, but generalisation beyond the trained demand is not established here. Three families (the powered base-model factorial, the exploratory sweep, and the trust battery) do not use the disjoint set, for the reasons given in Appendix B.8; every other closed-loop *controller* number in §6 is a disjoint-seed number.

5.4 Baselines and controls

Fixed-time is the floor every deployed junction achieves: each network’s own default cyclic plan, the same phase menu as every other arm. **MaxPressure** is the theoretically-grounded adaptive baseline [27], and doubles as the shield’s fallback policy. The control for every fine-tuning comparison is the *same base model compiled through the identical int4 pipeline* (denoted stock, or ft0), not the vendor catalogue artifact, for two reasons. I observed the catalogue Qwen3-0.6B GPU artifact emitting corrupted tokens after long service uptime, an unlogged operational observation rather than a measured finding (§6.12). Independently, and sufficient on its own, that artifact differs from my compiled one in quantisation recipe even when healthy. Holding the

compilation pipeline fixed leaves the fine-tuned weights as the single manipulated variable.

One model class is excluded by measurement rather than swept: phi-4-mini-reasoning’s p99 decision latency (11.1 s) exceeds the 10 s decision interval on this hardware, so no deployment could serve it in time. That is a deployability finding, not a control one, since the harness blocks on the model call with simulated time stopped and a closed-loop cell would flatter a slow model rather than penalise it; the p99 is one format of four on one seed, the other three at 8.69, 9.61 and 9.74 s, and the model is recorded as latency-excluded with its single corridor probe retained descriptively. Two further controls target specific hypotheses: a **leave-one-topology-out** (LOTO) student, trained with all Old Street states removed, tests whether distillation gains are topology memorisation, and four **format-specialist** students, each trained on one prompt format alone, test whether a deployed junction would prefer a per-format model over a generalist. The answer (§6.3) was no, at the single scale (3.8B) that sweep ran at.

5.5 Experiment inventory

Table 6 (Appendix B) maps every experiment family in §6 to its design and artifact trail; all raw cells, per-seed records and aggregation scripts are retained in the repository, so every number is recomputable. The campaign spanned three machines and roughly six weeks of wall-clock compute.

6 Results

Results follow the order the evidence was built: the baseline landscape, then the demand calibration the throughput guards forced and the headline result it yields, then authorship, forensics, audit, trust and reliability.

6.1 Baseline landscape: what the fixed-time floor revealed

In the exploratory sweep (three seeds per cell, daily-resolution demand) the theoretically throughput-optimal MaxPressure [27] appeared *worse than naive fixed-time on every real London topology* (Euston +15%, Old Street +5%, Bloomsbury +3%) yet dominated the synthetic grids (−27% on grid4×4, −19% on grid3×3). Replication kills it: Table 1 recomputes it paired per seed over the thirty training-disjoint evaluation seeds.

At $n=30$ the picture is a **sign flip that tracks geometry** (Figure 3), not a reversal: on Old Street, the study’s most complex geometry at five arms and seventeen movements, MaxPressure is *significantly better* than fixed-time, while on the oversaturated Bloomsbury grid the two are indistinguishable; calibrated to clear, the same network puts MaxPressure 8.8% ahead, so every “tied on Bloomsbury” statement is scoped to the oversaturated regime. Calibration does not settle the corridor, where the same rule leaves the comparison inconclusive ($p=.095$, a CI straddling zero, absolute

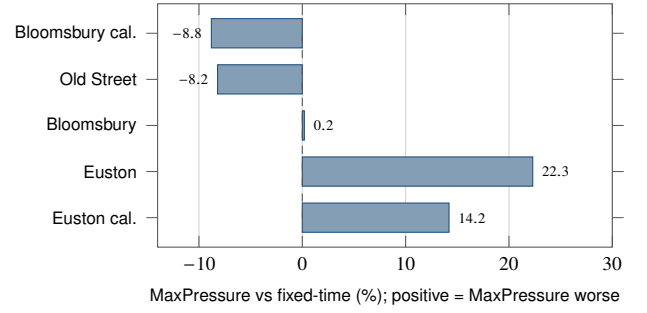


Figure 3: The classical baseline is not uniformly strong: MaxPressure is significantly worse than naive fixed-time on the Euston corridor, significantly better on Old Street, and tied on the oversaturated Bloomsbury grid; calibrated to clear, the grid favours MaxPressure and the corridor stays inconclusive. Every controller claim is therefore stated against the fixed-time floor with MaxPressure alongside.

means favouring MaxPressure by 2.4%),¹ so saturation is unconfirmed as a *sufficient* account of MaxPressure’s weakness: the networks differ in several respects at once and nothing here varies them independently. Before calibration MaxPressure is *significantly worse* than fixed-time on the saturated corridor too, with a caveat: its per-seed spread is heavily skewed, and on absolute rather than relative delays the paired test would not reach significance ($p=.083$), making Euston the least robust of the three real networks to how the question is posed.

A third classical baseline, longest-queue-first, which serves the largest upstream queue with no downstream term, is statistically indistinguishable from MaxPressure on all five topologies (−0.9% to +13.4%, every $p \geq .30$) and reproduces its pattern against the naive floor (Appendix B.9), so **the downstream term buys nothing measurable**; the intervals are wide, so this is a non-rejection rather than a demonstrated equivalence, and saturation, link storage and the assumed turning split all remain candidates.

MaxPressure’s advantage is uniform and substantial only on the regular synthetic grids the literature favours, which replication did not reach: those figures remain three-seed. Backpressure theory suggests why the arterial is its weakest case, a saturated short-link corridor violating both the unbounded-storage and heavy-load assumptions of §2.1 [7, 6]; that is a hypothesis consistent with the Euston direction, not a demonstrated explanation, since nothing here manipulates storage capacity. Every controller claim is therefore stated against the fixed-time floor, with MaxPressure alongside.

¹The +14.2% is the mean of per-seed ratios, the convention used throughout this section, and it is inflated by a single +140.3% seed. On absolute paired delays the same comparison gives $p=.82$. The row is reported because the convention is applied uniformly, not because it supports a direction.

Topology	MaxPressure (s)	fixed-time (s)	rel. (%)	95% CI	<i>p</i>
Euston peak-hour	263.8	233.0	22.3	[+5.6, +39.1]	.011
Bloomsbury grid	529.0	528.6	0.2	[−1.6, +2.0]	.82
Old Street	146.6	161.0	−8.2	[−12.3, −4.2]	2.6×10^{-4}
Bloomsbury calibrated	158.4	173.7	−8.8	[−9.2, −8.3]	8×10^{-27}
Euston calibrated	151.7	155.4	14.2	[−2.6, +31.0]	.095

Table 1: MaxPressure vs fixed-time, paired per seed ($n=30$). Positive = MaxPressure worse. CI is 95% on the mean relative change and p is the one-sample t on that same quantity, the convention used throughout this section.

6.2 Base-model factorial

The base-model sweep (eight servable catalogue models $\times$ four prompt formats $\times$ six topologies) established three facts before any fine-tuning. It is a sweep, not a full factorial. Three models were never swept on the later measured-demand corridor, phi-4-mini-reasoning is latency-excluded (§5.4) after a single-seed corridor probe, and the ANOVA is fitted on the balanced subset; the 6 GB card caps the ladder at ~4B (§4.2).

The sweep ran three seeds per cell with MaxPressure as its only comparator. In it the best shielded stock arm beat MaxPressure on every topology, from −2.4% on grid4x4 to −31.0% on the daily-demand corridor (Bloomsbury −6.1%, Old Street −15.4%). Those are per-map bests. Averaged over all stock arms the sweep is far worse than MaxPressure on the synthetic grids (+20% and +36%) and mixed on the real maps, from −15.8% on the daily-demand corridor to +0.1% on Euston peak-hour, and the per-map win does *not* survive replication on the measured-demand corridor. It motivated the powered design rather than counting as evidence. Results must also be conditioned on topology, because pooling across maps is a Simpson’s-paradox trap. And per-decision reasoning quality does not predict closed-loop value across base models. On 80 identical junction states cloud frontier models agree with the delay-optimal heuristic at 92–96% against 51–72% for local SLMs, yet a near-chance local model still won closed-loop cells; within one base it can (§6.5). The controller’s value is plausibly systemic, SLM, shield and gating together, though no arm removes the shield or the gate to test that, and this battery is open-loop by construction.

An explanatory two-way ANOVA (2 models $\times$ 4 topologies $\times$ 3 seeds, config fixed) confirms the structure, topology dominating (partial $\eta^2=0.97$), the model mattering (partial $\eta^2=0.83$) and the two interacting strongly ($F=32.2$, $p<0.001$, partial $\eta^2=0.86$), though at three seeds per cell this is exploratory, not the powered design.

The powered stock-model replication ($n=30$ per cell, seeds 1–30, four catalogue models $\times$ two formats, both baselines) sharpens this per topology; stock models have no training set, so training-disjoint seeds are not required. On **Old Street** stock SLM control achieves the campaign’s strongest powered wins against the fixed-time floor: Qwen2.5-0.5B sota −11.8% ($p<0.001$), Qwen3-0.6B sota −9.6% ($p<0.001$), Phi-4-mini myopic −8.6% ($p<0.001$),

one arm also significantly beating MaxPressure (Qwen2.5-0.5B sota, −4.9%, $p=.028$).

On **Euston peak-hour**, with demand set to the DfT-measured peak-hour magnitude, directional split and vehicle mix for the A501 count points (turning proportions assumed, single weekday survey), the result reverses. Every stock arm is **worse than the fixed-time floor by 7.4% to 19.7%**, significantly on three of eight arms (two-sided p .017–.026, a fourth at $p=.053$), yet statistically indistinguishable from MaxPressure. That is not a null but a measured harm. The measured magnitude loads the corridor *less* heavily than the daily average it replaces (780 vehicles against 1,071, 45.3% completing against 33.1%). Whether the lighter load is what makes the harm visible cannot be said here, because reducing Euston’s saturation further does not behave as a saturation-compression account predicts (§6.5). **Distillation does not repair this:** the fine-tuned Qwen and Phi arms are +15.5% and +16.2% against the same comparator (Table 2), inside the stock band. An earlier draft claimed the opposite, an artefact of scoring fine-tuned arms against MaxPressure and stock arms against fixed-time. Whether the deficit is topological or a residue of congestion that calibration could not remove is left open in §6.5.

The replication also yielded an unplanned methodological finding. On the 26 seeds shared between runs, the *catalogue* Qwen3-0.6B artifact and my identically-prompted, identically-shielded compile of the same weights differ by nine points of mean relative delay (+4.1% vs −5.2% against MaxPressure) on this high-variance topology. Quantisation and packaging provenance alone shift the headline sign (Appendix B.10). Hence the identical-pipeline control of §5.4: “the same model” is not the same controller across serving artifacts.

6.3 Offline distillation: saturation at 0.6B

Table 7 (Appendix B) gives the complete offline campaign on the frozen 5,580-state holdout as served (WebGPU, int4): both student scales, stock versus fine-tuned through the identical int4 pipeline, plus LOTO and format-specialist controls; Figure 4 plots the four main rows.

Two Qwen checkpoints appear in this report: the tabled row is the *generalist* student, trained and evaluated on all four formats, while every closed-loop result uses the *sota-only* student, trained on the delay-aware format alone and served in it, which scores 97.4% against the generalist’s 97.7%



Figure 4: Offline decision accuracy against frozen teacher labels, int4 as served. Distillation closes almost the whole gap and the two fine-tuned students land on each other at every format; the myopic exception at 88% is the privileged-teacher ceiling (§6.3).

on the only format either is asked to act in. The offline table therefore characterises the distillation rather than the provenance of the closed-loop arms. Five findings follow.

1. **Scale saturation.** The 0.6B and 3.8B generalists are statistically indistinguishable on every format: the largest gap, 0.8 points on myopic, sits against a standard error of about 1.2 points on $n=1,510$, so no two-proportion test approaches significance. That test is conservative, since both students are scored on the same frozen items, but only aggregate accuracies per format were retained, making the exact paired test (McNemar) uncomputable after the fact. Both sit at the label-noise ceiling on three of four formats (97 to 99% against 97.3 to 99.3% agreement between the two frontier models that produced the labels, on 600 cross-checked items), myopic excepted at 88%. Six times the parameters buy no measurable per-decision gain.
2. **Where the gain lands depends on the base.** The 0.6B is lifted by +29.7 to +42.6 points on every format; Phi-4-mini, already 100% format-compliant, gains most on the delay-aware sota format (+38.1) and negligibly on myopic (+1.2), so for a capable base fine-tuning raises decision quality from an already high floor (87.5, 59.0, 86.0, 87.0) rather than instilling format discipline.
3. **Generalists only.** No format specialist beats the generalist on its own format by more than 0.7 points while off-format costs run from 0.8 to 9.2 points, and the myopic specialist loses *even on-format* (87.0 vs 88.7), consistent with mixed-format training regularising against noisy privileged-teacher labels; the sweep ran at 3.8B only, and the sota-only Qwen student is not part of it.
4. **No detectable topology memorisation.** The LOTO student, trained with all Old Street states removed, matches the generalist on every format; Old Street is 4.2% of the sota rows and 0.4% of coordination, so this bounds a held-out-topology deficit near 7 points rather than measuring one, and per-topology accuracy was not retained.
5. **Quantised serving costs no measurable accuracy here.** The CPU and WebGPU int4 artifacts score identically in aggregate (97.42% both on the sota holdout, $n=1,510$).

Per-item outputs were not retained, so this is equality of accuracy, not demonstrated output identity, and the “lossless” claim is scoped narrowly in §7.3.

The myopic ceiling (~88%) is most likely intrinsic to the format rather than a training failure: the teacher was privileged with waiting-time state the myopic prompt hides, an Orca-style label-only distillation cost [18], and every myopic arm bar the stock 0.6B lands within 1.7 points of that ceiling, consistent with a label-noise floor; the non-privileged-label control that would establish it was not run.

6.4 Closed-loop outcome: delay decouples from accuracy

Table 2 reports the Qwen3-0.6B closed-loop sweep: fine-tuned (ft) versus identically-compiled stock, three topologies $\times$ 30 training-disjoint seeds, paired per seed against *both* classical comparators (§6.1).

The comparator decides the outcome. Against the fixed-time floor every arm significantly beats it on Old Street and none does on the other two maps; against MaxPressure the only arm in this table to reach significance is stock on the Bloomsbury grid (+2.6%, $p=1.3\times 10^{-3}$), a harm and not a gain, though one catalogue stock arm does beat MaxPressure on Old Street (§6.2), and the calibrated regime changes both pictures (§6.5). Comparator-invariant is the negative that matters: a +42-point offline accuracy gain produces no consistent delay change in these three regimes, and the pooled effect is indistinguishable from zero under either comparator, the **delay-decoupling paradox** the decision battery anticipated. Fine-tuning’s one significant effect is protective: stock control significantly *harms* the congested Bloomsbury grid where the fine-tuned controller is neutral, the head-to-head favouring fine-tuning by -1.8% ($p=.022$). §7.2 takes up the mechanism: masking confines both arms to one conflict-free menu and the anti-starvation floor serves 23–61% of ticks, though the floor’s share varies with the arm (§6.6), so much served behaviour is identical by construction whatever the proposal quality. I ran no unmasked arm, so this is a design-level reading, not a measurement, and the headroom may be small to begin with: external calibration

Topology	Arm	vs fixed-time	vs MaxPr.	auth	a-s	acc	div
Euston peak-hour	ft	+15.5%	−3.8%	0.64	0.36	0.70	0.12
	stock	+7.6%	−6.2%	0.62	0.38	0.68	0.46
Bloomsbury grid	ft	+0.8%	+0.7%	0.70	0.29	0.73	0.17
	stock	+2.8%	+2.6%	0.62	0.38	0.65	0.46
Old Street	ft	−11.6%	−3.1%	0.56	0.44	0.90	0.11
	stock	−9.7%	−1.1%	0.39	0.61	0.54	0.60
Pooled ($n=90$)	ft	+1.6%	−2.1%				
	stock	+0.2%	−1.6%				
Euston	phi ft	+16.2%	−3.8%	0.64	0.36	0.70	0.13
Bloomsbury	phi ft	+0.8%	+0.6%	0.70	0.30	0.73	0.17
Old Street	phi ft	−9.7%	−1.0%	0.57	0.43	0.88	0.12
Pooled ($n=90$)	phi ft	+2.4%	−1.4%				

Table 2: Closed-loop delay against both classical comparators and decision composition ($n=30$ disjoint seeds per cell); negative Δ favours the controller; bold marks a significant improvement over the fixed-time floor ($p < .05$, full tests in Table 8). *auth*: SLM-authored share of actuations; *a-s*: anti-starvation share; *acc*: proposal acceptance; *div*: divergence from MaxPressure among SLM-authored decisions.

puts the whole LLM-vs-MaxPressure gap at 1–2% [32].

Because paired t -tests on means hide the skew the catastrophic tail of §6.7 puts into the per-seed distributions, Table 8 (Appendix B) adds 95% CIs and a Wilcoxon signed-rank test on the *typical* seed. **The Old Street win holds on the typical seed** as well as the mean ($p \leq .002$ both tests); the Euston intervals are wide and zero-straddling on the topology with the heaviest failure tail, so the characterisation is *typical-case behaviour with a catastrophic tail*, pointing at tail control (a per-decision timeout, or an outlier detector reverting to the classical policy) rather than at better average decisions. The rank-test results are not confirmatory and would not survive a Holm correction across the roughly forty paired comparisons in this report; the fine-tuned-versus-stock head-to-head, the study’s primary contrast by design, does (Appendix B.11).

The Phi-4-mini confirmation arm (final rows of Table 2) extends saturation end-to-end: the 3.8B fine-tuned student’s closed-loop profile is not distinguishable from the 0.6B’s on either comparator, with the same divergence band (0.11 to 0.17) and authorship profile, and on the calibrated grid the intervals bound any difference within $\pm 0.38\%$ of delay; it serves at the lowest latency of my compiled arms (Table 10, Appendix B). On the scale-threshold question that is a bounded non-difference, not a demonstrated equivalence (§6.5).

6.5 Throughput guards, and the result the gridlock was hiding

Mean delay can be improved by stranding vehicles, so §5.1 commits to guard metrics alongside it. Table 3 changes how two of the three topologies read.

Of the three original cells, only Old Street is measured in a free-flowing regime. It clears, teleports are effectively absent, mean delay is well-posed, and every arm significantly beats the fixed-time floor there (Table 8). Euston sits between,

at 45–51% completion. There the SLM arms *complete more vehicles than MaxPressure* (380 and 392 against 353) without a significant delay advantage, though not more than the fixed-time floor’s 399; no significance test was run on completion. **Uncalibrated Bloomsbury is a gridlock-regime measurement.** Under 15% complete and over a hundred teleport out of deadlock, so its mean delay compares two congested-failure modes. The stock arm completes fewer vehicles and teleports more than MaxPressure, so the stock harm survives the guard test in direction, but it is regime-scoped. Completion also reproduces §6.1’s ordering without using delay, the tie appearing as MaxPressure marginally worse (14.6% against 15.1%). So **the sign flip is not an artefact of a fragile mean**, though completion is as consistent with saturation or link storage as with geometry.

Bloomsbury carried the study’s only significant closed-loop result against MaxPressure, so I re-examined it. Tripling the horizon to 3,600 steps raised completion only to 35% while multiplying teleports roughly sevenfold (121 to 834): the network was over capacity, not under-simulated. Subsampled demand located the boundary sharply, clearing at 40% (87.6% completion, zero teleports) and collapsing again at 55% (57.5%), so I added a demand-calibrated variant at 40% (686 vehicles) and re-ran all three arms over the identical thirty seeds, retaining the oversaturated cells. Whether this is a principled correction or a search for a more favourable test bed is argued in Appendix B.12: the guard metrics predate the decision to look, the 40% target matched Old Street’s completion rate rather than any effect size, and the rule was applied uniformly to Euston peak-hour, which also fails the guard and is where SLM control looks *worst* (§6.2), so calibrating it risked the conclusion rather than protecting it.

Calibrating Euston does not rescue the corridor, and the distillation effect does not survive there at full size. Completion rises from 45–51% to 68–74%, but in the

Topology	Arm	Loaded	Compl.	Teleports	Undep.
Euston peak-hour	fixed-time	780	51.1%	22.2	184
	MaxPressure	780	45.3%	31.1	218
	qwen ft	780	48.7%	28.1	200
	qwen stock	780	50.3%	23.3	199
Bloomsbury grid	fixed-time	1715	15.1%	115.0	328
	MaxPressure	1715	14.6%	119.3	336
	qwen ft	1715	14.2%	120.0	341
	qwen stock	1715	13.5%	135.6	348
Old Street	fixed-time	480	83.3%	0.7	1
	MaxPressure	480	85.8%	1.3	0
	qwen ft	480	86.7%	0.5	0
	qwen stock	480	84.4%	1.0	1

Table 3: Guard metrics over the same $n=30$ training-disjoint seeds as every delay comparison. Compl. is completed vehicles as a share of those loaded within the 1,200-step horizon; teleports are SUMO’s gridlock-escape removals.

MaxPressure arm 7.7 vehicles per run still teleport and 56.9 never depart, whereas on the calibrated grid both classical comparators record 0.0 teleports and 0.0 undeparted and only the stock arm differs (1.70 teleports, 5.7 undeparted). On the partly cleared corridor the fine-tuned arm is still +8.5% against the fixed-time floor ($[-2.4, +19.5]$, $p=.12$) and the stock arm +34.8% ($[+5.3, +64.4]$, $p=.023$). The mean spares the fine-tuned arm only because the tail is enormous: it beats the floor on six of thirty seeds, its median seed is +11.7%, and the rank test rejects ($p=.016$). Neither arm differs significantly from MaxPressure (-0.3% at $p=.93$, +21.9% at $p=.062$). The head-to-head favours fine-tuning in direction and on the typical seed (median -7.4% , 23 of 30 seeds, Wilcoxon $p=.045$), but its mean is -3.7% and not significant ($[-19.1, +11.7]$, $p=.63$), across per-seed outcomes from -71.6% to $+104.5\%$.

Two conclusions follow, the first against me: **saturation explains the Bloomsbury result; on Euston it is reduced, not tested**. The same correction yields on the corridor a fifth of the grid’s magnitude and no significant mean, though in absolute delay (-27.4 s against -35.9 s) the two are comparable. I cannot attribute the surviving deficit to topology, since nothing here varies geometry while holding demand and storage fixed. Topology and residual saturation both stay live, and the grid effect did not replicate on the second network I calibrated. The other conclusion concerns method. The threshold was applied where it could damage the conclusion, and did: the rule, take the largest demand fraction that still clears, produced an unfavourable result, reported as such.

The gridlock was masking the effect, not creating it (Table 4, Figure 5). On the calibrated network the stock controller degrades mean delay by **11.9% against the fixed-time floor** and **22.7% against MaxPressure**, an order of magnitude beyond the +2.8% it costs oversaturated, while both fine-tuned students **beat the fixed-time floor by 8.8%**, within 0.03% of MaxPressure. The head-to-head is the largest, most certain effect in this report: **18.3% less delay**

than the identically-compiled stock model ($[-19.8, -16.8]$, $p=2.0\times 10^{-21}$), **winning 30 of 30 seeds**, the Phi student reproducing it (-18.3% , 30/30, $p=8.2\times 10^{-22}$). The thirty seeds vary SUMO’s microscopic noise over one fixed 40% demand draw, so they replicate within a scenario rather than across demand; the mechanism is the one §7.2 develops. Throughput corroborates it (stock 76.4% against the fine-tuned 87.4%), and the stock arm shows no serving pathology (484 valid proposals of 546 decisions, no fallbacks, normal latency), so the degradation is in the decisions, not the endpoint.

This also bounds the scale-saturation result instead of leaving it a non-rejection: the 3.8B and 0.6B students differ by +0.03% with a 95% CI of $[-0.32, +0.38]$, so any difference between the scales is within $\pm 0.38\%$ of delay. That stops short of demonstrated equivalence, which would need a margin pre-registered against an operational threshold (a TOST procedure) this study did not define.

An oversaturated test network can hide a large effect: the original Bloomsbury cells alone would have concluded a marginal 1.8% at $p=.022$. On Euston the same correction did not reveal a comparable effect, so this is a demonstrated failure mode on one network of two, not a systematic bias whose size I can quantify. The literature evaluates predominantly on synthetic grids under demand rarely reported in clearance terms (§2.2), so saturation-driven compression of published effect sizes deserves attention; reporting completion and teleport counts alongside delay would expose the regime, though not the compression, since grid3×3 is oversaturated at 37.8% completion and still shows -19.1% .

6.6 Authorship: what fine-tuning actually moves

The structural columns of Table 2 carry the report’s central finding. It emerges only if the three denominators of §3.3 are kept apart (full composition per cell: Table 9, Appendix B).

Any claim that an SLM arm “matches” MaxPressure would be circular were the shield quietly substituting the classical phase. **It is not**. Across fourteen arm-by-topology

Regime	Arm	Δ vs fixed-time	Δ vs MaxPressure	Compl.	Auth.
Oversaturated (14% clears)	qwen ft	+0.8 [-1.0, +2.6]	+0.7 [-0.7, +2.1]	14.2%	0.70
	qwen stock	+2.8 [+0.9, +4.6]	+2.6 [+1.1, +4.1]	13.5%	0.62
Calibrated (87% clears)	qwen ft	-8.8 [-9.4, -8.2]	+0.00 [-0.56, +0.56]	87.4%	0.76
	phi ft	-8.8 [-9.3, -8.3]	+0.03 [-0.43, +0.48]	87.5%	0.77
	qwen stock	+11.9 [+10.0, +13.8]	+22.7 [+20.6, +24.7]	76.4%	0.59

Table 4: The same controllers under two demand regimes on the same Bloomsbury network ($n=30$ paired seeds each); positive Δ is worse than the fixed-time floor.

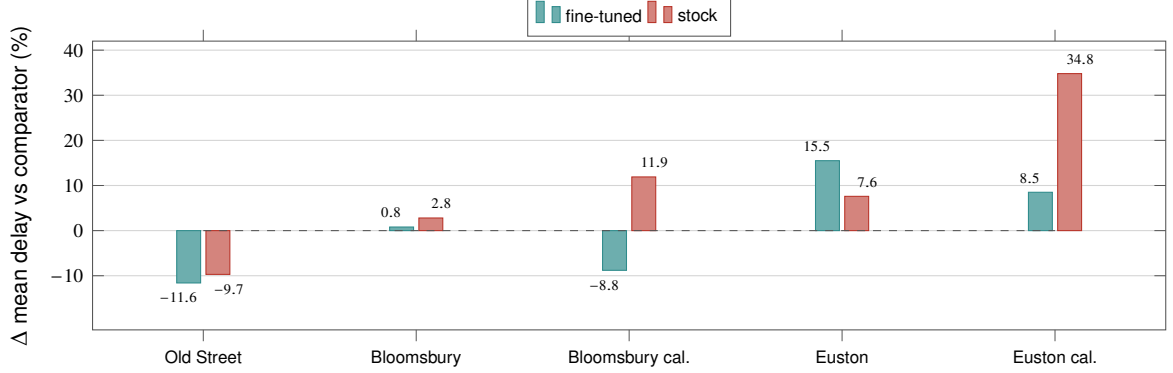


Figure 5: Closed-loop delay against the fixed-time floor, $n=30$ training-disjoint seeds per cell (negative is better). The distillation effect is the gap between the bars: largest on the calibrated Bloomsbury grid, not reproduced at that size on the calibrated corridor.

cells and 420 seed-runs the substitution path fired in exactly one cell, the fine-tuned arm on the Bloomsbury grid, at 0.54% of decisions on average (110 in total, 8.99% on its worst single seed). Every other cell is 0.00% on every seed (shield_fallbacks_none is zero in every SLM cell).

The anti-starvation floor is the large non-SLM component, at 23 to 61% of ticks, and it authors 24 to 43% of the *MaxPressure* arm on the same maps. The rule is identical in every arm but its share is not. Paired per seed, the stock arm carries 4 to 18 points more floor than *MaxPressure* on every map ($p \leq 1.2 \times 10^{-9}$), while the fine-tuned arm sits at the *MaxPressure* rate. So the floor intervenes most in the arm that performs worst, and what fine-tuning removes is the stock proposer’s excess over the no-SLM rate.

Fine-tuning raises the SLM’s true authorship of actuations, most where stock was weakest: Old Street 0.39 to 0.56, Bloomsbury 0.62 to 0.70 (0.76 once calibrated), Euston 0.62 to 0.64 (Figure 6). The floor column is the arithmetic complement (Old Street 0.61 $\rightarrow$ 0.44), so it adds no independent evidence. Proposal acceptance, which does, rises with it (0.54 $\rightarrow$ 0.90). The natural reading is that the distilled model serves neglected approaches before the fairness timer must. That cannot be separated from the arm simply tripping the timer less often, because the logs record only who served each tick, not which approach. The honest headline is the authored share, not the acceptance rate: even the best fine-tuned arm leaves 23 to 44% of actuations authored by a hard-coded timer.

Cutting against the simple reading, **the fine-tuned arms**

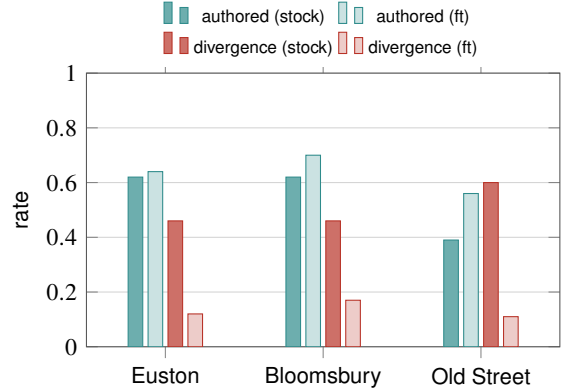


Figure 6: What fine-tuning moves ($n=30$ seeds per bar). Teal: the SLM’s share of *logged decision ticks*. Red: divergence from *MaxPressure* among SLM-authored decisions.

land level with *MaxPressure* while they choose what *MaxPressure* would have chosen. Divergence among SLM-authored decisions falls from 0.46–0.60 to 0.11–0.17, and on the calibrated networks that carry the headline it falls to 0.04 and 0.07, the two lowest in the study, with authorship at 0.59 against 0.76–0.77. That is genuine authorship of a near-classical policy. But the deflationary reading is strongest exactly where the delay result is largest: distillation may have taught the student to imitate the classical reference, an explanation neither established nor excluded. Two observations bound it. The teacher was a frontier model reasoning over accumulated waiting, not *MaxPressure*, and the students agree with its held-out labels at 97–99% (§6.3), so

any convergence is at most indirect. And closed-loop delay cannot discriminate imitation from rediscovery: against the fixed-time comparator MaxPressure is +22.3% on Euston peak-hour against the fine-tuned arm's +15.5%, and on the calibrated grid the two coincide within 0.03%. Only the remaining one-in-ten divergences, where the arms differ from both baselines, could discriminate them. I have not characterised those, so the question needs a decision-level counterfactual study I did not run. What survives is the audit-relevant claim at its true magnitude: over the stock Old Street deployment an accident audit could attribute 39% of actuations to the learned component, over the fine-tuned one 56%, the balance falling to a deterministic timer. That is a smaller claim than *the model is the controller*.

6.7 Failure forensics: the audit layer explains the tail

The audit architecture's value shows most clearly when control fails. Of the 270 uncalibrated-topology closed-loop cells, 14 (5.2%) are catastrophic (delay $\geq +20\%$ vs the seed-matched MaxPressure baseline), and all 14 signed chains verify, so each failure has a provable record. **Failures are a controller property, not seed hardness.** The catastrophic seeds are baseline-easy or median demand realisations (baseline delay ranks 1–20 of ~ 35 , never the hard tail), and no seed breaks all three arms. The two failure regimes also have different, attributable mechanisms. Four of the stock arm's five failures are its own served decisions. In the worst cell (Euston s13, +110.3%) the chain shows 214 of 335 decisions SLM-authored, 119 divergent proposals served (override 0.556), zero shield fallbacks and normal anti-starvation, so the collapse was the model's policy; same-seed fine-tuned arms sit at +1.1% and +5.5%. The fifth (Euston s19) sits at arm norms and the chain clears it. And **the two fine-tuned students converge at the trajectory level on a substantial minority of cells.** Of the 90 shared cells, nine produce *bit-identical* mean delay, 25 more agree within 0.5 points, and 56 differ, and the students' catastrophic cells are paired: the 0.6B student's four all coincide, seed for seed, with four of the 3.8B student's five (Appendix B.13). I read it as the behavioural counterpart of offline saturation. Partial convergence is equally consistent with both students defaulting alike on easier cells, and I have not separated those explanations.

Across the calibrated regimes the same catastrophic cells show the clearest effect fine-tuning has anywhere in this study, and it is not an effect on mean delay. Paired over the same thirty seeds against the identically-compiled stock control, catastrophic cells fall from **20 to 0** on the calibrated Bloomsbury grid and from **8 to 1** on the calibrated Euston corridor, the 3.8B student reproducing the grid result at 0. Where the network cannot clear the tail does not move: 3 to 3 on Euston peak-hour, 2 to 1 on Old Street, 0 to 0 on the gridlocked grid. **Distillation removes the harm tail on both networks that clear**, including the corridor

where it moved no mean delay. The tail and the mean answer to the same clearance condition but not to the same degree, and I did not isolate the mechanism. One honest gap: this campaign persisted aggregate decision statistics per cell, not per-step audit records, so *when* within a run the divergence concentrated is unrecoverable here, though the per-step chain is retained in the dedicated accident-suite runs (§6.8). Full tables in `results/failure_forensics.md` (Appendix B).

6.8 Audit and crash reconstruction

The audit layer was evaluated on seven pre-defined collision scenarios spanning the legally distinct cases (red-light running, EV exemption, maintenance vehicle, amber gambling, conflicting green, dark signal, spoofed EV). All seven prove end-to-end: hash chain verified, signatures verified, and an imposter re-signing attack caught in every scenario. The governing UK rule is *inferred deterministically from the verified facts*: spoofed-EV correctly denies the exemption and anchors fault on the driver, and conflicting green alone attributes (weak, misfeasance-only) authority fault. The output is a cited evidence pack for a human adjudicator, never a verdict [3]. Above the per-run chain, the Certificate-Transparency-style completeness layer of §2.6 passes its property battery over a twelve-entry batch, including detection of a witness equivocation, with two limits: the cross-auditor's own honesty is an **unverified assumption**, and the batch is **not externally anchored** (Appendix B.14).

6.8.1 A worked evidence pack

Figure 7 traces the hardest of the seven end to end, a collision at a junction that emitted a conflicting green. Each cited rule carries the fact that triggered it, so an investigator can audit the inference rather than read a conclusion, and the doctrine cited runs against the deployer.

6.9 Trust-coefficient battery

The mechanism passes its 9/9 property battery (§3.5). In closed-loop traffic I ran it against fine-tuned 3.8B, stock 0.6B and fine-tuned 0.6B controllers, at three seeds per cell and with gating off, so the figures are descriptive and not comparable cell-for-cell with Table 2. Appendix B.15 gives the detail. Three findings follow. **Detection is controller-independent by design and in measurement**: every controller catches and collapses a blatant liar where lies are injectable, while a calibrated stealth attacker evades essentially everywhere, an open limitation. **The delay cost of blatant lying survives detection**, adding +2.7 to +11.5 points on the synthetic grids, because detection bounds future influence without undoing congestion. And **stealth's harm is positional, not congestive**. The threat model for a fine-tuned deployment is therefore asymmetric, and that is the deployment that matters, since a controller acting on its coordination inputs

```

VERIFIED RECORD (seq 0, junction J, hash 93dc0a0b..., t=100.0)
  chain verified: True   signatures verified: True
  imposter re-sign attempt: CAUGHT
VERIFIED FACTS
  signal_state = green      conflicting_green = True
  crossing_class = civilian second_party_on_green = False
  key_present = False      corroborated = False
GOVERNING RULES (inferred from the facts above)
  LR-authority-misfeasance [Bird v Pearce] fault: low (common law)
    "the audit shows the system emitted a conflicting green
    (a positively-wrong indication) -- misfeasance, not omission"
  LR-authority-nonfeasance [Gorringe v Calderdale] fault: low (common law)
    "authority fault otherwise defaults to zero; it turns non-zero only
    on this misfeasance/conflicting-green branch (low confidence;
    legal_causation NOT_ASSESSED)"
OUTPUT: cited evidence pack for a human adjudicator -- not an adjudication

```

Figure 7: The conflicting-green evidence pack, reproduced from `results/experiment_crash_law.json` (record, facts, rule ids, fault weights and both `why` strings verbatim; headers and closing line editorial): the only scenario of the seven in which authority fault is non-zero, the system’s own record used against its operator.

is what poisoned inputs can move (§6.6). Distributional detection is the right next defence (§7.4).

6.10 Emergency preemption: what the priority path is worth

The authorisation machinery serves a concrete function, measured over thirty seeds on the synthetic arterial corridor with a scripted ambulance and spoofing attack in three modes, no preemption, naive and defended (Appendix B.16, Table 11). All four intervals exclude zero: **signal priority is worth about half a minute to an emergency vehicle** (31.1 s), **security has a measurable price** (8.0 s to the ambulance), and **that price buys quantified protection** (4.6 s mean and 16.1 s worst-case cross-traffic harm avoided under attack). These are per-event magnitudes from one scripted ambulance and one attack window per run, not expected values under a naturalistic call rate, and the baseline characterises no specific deployed system.

6.11 The fair re-test of the SLM authorisation negative

An early measured negative, the raw SLM losing to a deterministic template on its legal-note job (citation-F1 0.12 retrieval-free and 0.34 with retrieval against the template’s 0.48 over 97 held-out notes), carried a supervisor-posed objection (Stott): the model was guessing because the policy was never in front of it. The re-test partially vindicates it (balanced 26-claim authorisation dataset, four models × policy-scaffolded and bare arms, scored against the deterministic knowledge-base gate; Appendix B.17). Stock Phi-4-mini rises from 42.3% to 69.2% class accuracy and its false-negative rate falls to zero. Even that best cell makes **four dangerous false grants** (four of 18 claims that must be denied) where the gate makes none in 26/26, so policy grounding rescues recall, not safety. The *fine-tuned* students are far worse than their own stock bases, ft-Phi falling to 19.2% scaffolded with 15 of 26 responses unparseable and ft-Qwen (the generalist of §6.3) unable to emit the required

schema at all bare (26/26 parse failures); format capture by its training schema is the natural reading, an inference and not a measurement. The retention probe’s mild MMLU decline (§7.3) understates the cost on near-neighbour structured tasks.

6.12 Serving reliability at the edge

Sustained on-device serving produced three documented reliability findings (a disk-full crash at 5,500 of 5,580 evaluation rows, undocumented manifest synthesis for builder-compiled models, and malformed output from the catalogue Qwen3-0.6B GPU artifact after extended uptime, the unlogged observation behind the pipeline-compiled control of §5.4), each mitigated in harness (Appendix B.19). Decision latency (Table 10, Appendix B) is comfortable at the median and not guaranteed in the tail: p50 is 0.94–1.75 s against the 10 s decision interval, vendor catalogue builds are faster still (0.68 s for the student’s own weights, the 2.6× compile penalty of §4.2 seen from the other side), but one 0.6B call took 13.79 s, the criterion that excludes phi-4-mini-reasoning. A deployment would need a hard timeout with deterministic fallback, which the architecture already supports.

7 Discussion

7.1 Answering the research questions

RQ1 (edge competence): qualified, and topology-dependent. At powered replication ($n=30$ disjoint seeds) the picture splits by map. Stock SLM control significantly beats the floor on Old Street. It is significantly *worse* on three of eight arms on the measured-demand Euston corridor, overturning a win the exploratory sweep had reported, and fine-tuning does not remove that deficit (point estimates +7.6% stock and +15.5% fine-tuned against the floor, neither significant). The defensible claim is narrow: competent on complex junction geometry, harmful on a saturated arterial,

and superior to the naive fixed-time plan only where the network can clear.

RQ2 (what distillation buys): delay where the network can flow, authorship everywhere, and it saturates at 0.6B. Offline, distillation is dramatic (45–69% → 88–99%) and scale-flat. On a demand-calibrated grid, on 30 of 30 seeds against the identically-compiled stock model, it lifts the controller above the naive fixed-time floor and is worth 18.3% delay. The two student scales sit within $\pm 0.38\%$ of each other, so scale saturation here is a bounded difference rather than an absence of evidence. Calibrating the Euston corridor by the same rule leaves the head-to-head not significant: the corridor deficit survives calibration, and I cannot attribute it to topology (§6.5). Independently of delay, distillation raises *ownership*: the SLM’s share of logged decision ticks rises from an already substantial 0.39–0.62 to 0.56–0.77. For a shielded, audited controller I argue ownership is the criterion that generalises, since it holds even where delay effects are compressed (§7.2). The saturation runs deeper than the accuracy tables show. The two students reproduce each other’s control trajectory exactly on a tenth of all cells (§6.7), though 56 of the 90 cells still differ and no ensemble was built or costed. Nor is distillation free: on an adjacent structured task the fine-tuned students are worse than their own stock bases, one unable to emit the required output schema at all (§6.11). That is acceptable for a component doing one decision type behind a deterministic gate, and not acceptable if the same weights were expected to serve the authorisation or legal-note roles.

RQ3 (provable audit): yes, as designed. Seven scenario reconstructions prove end-to-end with tampering caught (§6.8). The legally weak branches are weak *in the law*, faithfully represented rather than overstated.

RQ4 (trust): partial, and the partiality is structured. The mechanism’s properties hold under all three controllers tested, because the ledger scores claims against local sensing rather than model behaviour. At three seeds per cell these are descriptive magnitudes, and the three limits of §6.9 stand. The design-relevant tendency is that a blatant attack costs *more* against fine-tuned controllers than stock. That would mean authorship cuts both ways, a model that acts on its inputs being moved further by poisoned ones. It is a tendency because it is not uniform: of the six controller-map cells where lies are injectable, the fine-tuned cost is larger in four and smaller in two. A trust ledger is not a sufficient defence, and this study did not run a distilled deployment without one.

7.2 Why accuracy decouples from delay

Three mechanisms plausibly explain the delay-decoupling paradox, and they differ in how well this study evidences them. The least evidenced is that *the shield may compress the outcome distribution*. Masking confines every arm to the same conflict-free menu and the anti-starvation floor

serves 23–61% of ticks, intervening most in the weaker arm (§6.6), so the gap between a 59% and a 97% proposer may be far smaller served than raw. I ran no unmasked arm, and the substitution path fires on 0.54% of decisions in one cell and 0.00% elsewhere, so it bounds nothing: a design-level argument, not a measurement. The next is that *the headroom was small*. External calibration from a 13× larger tuned model puts the whole LLM-vs-MaxPressure delay gap at 1–2% [32], and my own decision battery showed a near-chance proposer winning closed-loop cells, because per-decision agreement with a delay-optimal heuristic is not the trajectory-level objective. The only one measured here is that *the test network must be able to express the difference*: on an oversaturated map every controller is compressed toward the same terrible mean, while the same experiment on a network that clears recovers the effect on every seed (§6.5). The latter two are properties of the problem and the test network; the first is a property of the controller I did not test. The practical reading: fine-tune to own the decisions, expect the delay gain and the removal of harm tails only where the network is not already saturated (§6.7), and never evaluate a controller on one that cannot clear.

7.3 Threats to validity

Five declared threats, with defences.

Label-only distillation (construct). I distil teacher labels, not reasoning traces. Orca-style trace distillation [18] might transfer more, and the myopic format’s ceiling prices the cost as a disclosed design property (§6.3).

Demand composition (external). Only the corridor’s vehicle counts come from measured DfT data; the turning proportions at each junction remain assumed. Pressure-based control acts directly on movement demand, so every controller comparison here is conditional on that assumption, and the corridor results are plausibly the most exposed to it, the A501’s short links making the through-versus-turning split decisive. Nothing in this study estimates that sensitivity.

Retention (internal). Aggressive SFT can erode general capability with scale-dependent forgetting [13]. A seeded 200-item MMLU sample through the same endpoints shows the 3.8B student at 57.5% (stock) against 53.0% (fine-tuned), consistent with mild forgetting and not significant at this sample size ($z=0.91$, $p=.37$); at 0.6B the probe is uninformative by floor effect, stock already scoring at chance (26.0%). No significant general-knowledge damage was detected, though 200 items bound only large effects, and that is not reassurance: the nearer-neighbour authorisation probe (§6.11) shows both students well below their stock bases, a narrowing onto the trained contract that general-knowledge benchmarks are poorly placed to detect.

Quantisation scope (construct). “Lossless” int4 is claimed strictly as equal aggregate holdout accuracy under weight-only RTN. Per-item outputs were not retained, so output identity is untested, and no general equivalence is

claimed, which the quantisation literature would not support [25, 5].

Simulation and demand realism (external). Everything is SUMO. Real-map demand *magnitude* is measured, but the peak-hour fraction, the temporal profile, the directional split and the vehicle-level realisations are assumed or synthetic; comparators exclude trained deep-RL systems. The audit scenarios use synthetic actors on real topology, so no real data subject exists; the two residual legal exposures a real deployment would face are set out in Appendix B.18.

7.4 Limitations and future work

Five directions follow from the findings. **A decision-level counterfactual:** replaying the divergent ticks and scoring what MaxPressure would have done against what the student did would give the imitation-versus-rediscovery question direct evidence (§6.6). **Formal shield synthesis** from a temporal-logic safety specification [1]: the per-junction product game may be small enough for exact synthesis, and the authorship metrics defined here would then carry minimal-interference guarantees. **Closed-loop objectives:** GRPO-style tuning against trajectory reward [32] attacks the decoupling directly, because label distillation optimises agreement rather than the trajectory, and moved delay on one network of five. **Stealth-robust trust:** detection calibrated to distributional implausibility rather than single-report bounds. **Hardware-in-the-loop:** the serving-reliability findings (§6.12) are the failure class a cabinet deployment must survive, and argue for a watchdog-supervised serving architecture.

7.4.1 Deployment vectors

The evaluation above measures the hardest case for this controller, a permanent junction already running the network’s own default signal plan, so matching it buys little. Two settings invert that arithmetic, temporary roadworks signals with no engineered plan and corridor-scale emergency-vehicle priority, and a September-2026 survey of published prices places the compute node at well under 1% of site cost; Appendix D gives the evidence, the open gaps and the cost figures, none of which originates from this study’s measurements.

8 Conclusion

Feasibility is a qualified yes, and the qualifications are the substance. Shielded on-device SLM control significantly beats the fixed-time floor on the study’s most complex junction geometry and, once distilled, on a demand-calibrated grid; the same controller is worse on a saturated arterial whether distilled or not, and elsewhere indistinguishable from that floor except on the oversaturated grid, where the undistilled controller is significantly worse. The classical baseline changes sign with the network, and what looked like

uniform weakness on real geometry was a three-seed result that replication retracted. The whole loop, from frontier-teacher distillation through QLoRA training to int4 serving, fits on one consumer device, and there distillation saturates: a student six times larger lands on statistically indistinguishable decision accuracy, pointing at the distillation ceiling rather than model capacity.

The central finding reframes what fine-tuning is *for* in a shielded controller. On the maps that could not clear, the 97%-accurate student does not significantly beat the floor, and neither does anything else inside the shield’s envelope, though I ran no shield ablation to isolate that compression. Where the network *can* flow, fine-tuning moves delay itself, on every one of thirty seeds (§6.5); applying the same calibration to the arterial corridor did not reproduce it, and I report that as the failed uniformity test it is. Fine-tuning moves authorship on every topology I tested (§6.6), and the audit layer proves who decided what, so authorship is what makes that audit concern the certified model rather than its fallback.

The nulls bound the claim, each reported with its mechanism: the corridor harm, the stealth attacker the trust ledger misses, the privileged-teacher ceiling, and the three serving-reliability findings. Within them, an audited, shielded, on-device SLM junction controller is possible today, at 0.6B, on a desktop-class 6 GB GPU, with published figures for cabinet-class hardware suggesting but not establishing the same envelope (Appendix D). For a signal engineer the reading is narrow: deploy this where the network can still clear or where no engineered plan exists, not on a saturated arterial; calibrate a test network to real demand before believing any delay figure; and treat the audit trail, not the delay, as what certification would rest on. What this study settles is narrower and firmer: fine-tuning does not change the audit trail, which verifies identically in every arm; it raises the share of the record the model owns, and it moves delay only where the network can clear, and not on every network that can.

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A Reproducibility artifacts

A.1 Environment and version pins

Every measurement in this report was produced under the pins in Table 5. The Foundry Local version is pinned deliberately: custom-model serving behaviour (registration format, execution-provider options) is version-sensitive, and a mid-campaign upgrade would have invalidated cross-run comparability.

A.2 Seeds and dataset splits

Training-state seeds: {3, 7, 11, 29, 42, 57, 91, 123}. Junction states for distillation were sampled only from these. **Closed-loop evaluation seeds:** $\{1..30\} \setminus \{3, 7, 11, 29\} \cup \{31, 32, 33, 34\}$ gives thirty seeds disjoint from training (§5.3). **Offline holdout:** split by deterministic content hash (HOLDOUT_MOD=12) and frozen before the first training run, which leaves 5,580 held-out examples against 59,533 training examples, from 65,113 accepted renderings (6 delay-aware states rejected by the sanity verifier for serving an empty phase without a justifying note, removing their sota and myopic renderings). Per format: sota 16,982, myopic 16,982, prediction 18,596, coordination 12,553.

A.3 Prompt contract

All four formats share one output contract: exactly `{"phase": N}`. The *same rendering code serves training, offline evaluation, and closed-loop control*, so no format skew can flatter the offline numbers. The delay-aware (sota) system prompt is reproduced in part:

```
You control ONE traffic-signal junction
(this junction only; no neighbour
information). For each green phase you
are given: the number of waiting
(queued) vehicles, the TOTAL accumulated
waiting time of those vehicles in
seconds, and whether it is the phase
currently green. Choose the phase to
serve next to MINIMISE total vehicle
waiting time and unnecessary stops -- not
simply the longest queue.
```

The myopic format withholds accumulated waiting, prediction adds short-horizon arrival counts, and coordination adds neighbour release notes. Parsing tiers are as given in §3.2.

A.4 The compile-and-serve chain

Reproducing a served custom model requires the following steps in order, each established by failure during this project.

1. Train QLoRA adapters ($r=16$, NF4 double quantisation, fp16 compute, completion-only loss, family-correct chat template).
2. Merge adapters to a bf16 checkpoint.
3. Compile with the onnxruntime-genai model builder (`-p int4 -e cpu`) using the pinned transformers pairing, because newer environments fail on attention-module or loss-utility imports.
4. Patch `tokenizer_config.json`: `tokenizer_class` must be a value the runtime accepts (GPT2Tokenizer), because transformers-5 era configs emit a backend name that fails to load.
5. For Phi checkpoints, strip remote-code `auto_map` entries.
6. **Synthesise** `inference_model.json` (model name plus prompt template). The builder emits none, and without it the model registers but returns HTTP 400 on every request.
7. Copy into the Foundry Local cache directory and, for GPU serving, set `provider_options` to WebGPU in `genai_config.json`.

A.5 Harness invariants

Three properties of the experiment harness underwrite the results. *Fail-loud*: API errors are counted and abort a run rather than being scored as answers. An unservable model records a skip, never a score. *Resumable*: cells are keyed by (model, config, topology, seed) and skipped if already measured, so a crash loses at most one cell. One zero-byte cell caused by a mid-write power loss was detected by JSON validation and re-run. *Recoverable*: after two observed mid-run service self-restarts, the offline evaluator rediscovers the endpoint, reloads the model, and retries failed rows with statistics intact.

B Extended results

B.1 Experiment inventory

Every experiment family in §6, with its design and primary output.

B.2 Per-topology closed-loop detail

The per-seed cells behind the means and paired tests of §6 are kept in `results/frontier_raw/`, one file per (model, config, topology, seed) containing the MaxPressure baseline delay, controller delay, relative change, decision composition (SLM / substitution / anti-starvation), from which authored share, acceptance and divergence derive, plus latency distribution and audit-chain check. The fixed-time floor is

Component	Pin
Serving runtime	Microsoft Foundry Local 0.8.119
Compile path	onnxruntime-genai model builder, int4 RTN, weight-only
Execution providers	CPU and WebGPU (equal aggregate holdout accuracy, §4.2)
Simulator	SUMO with TraCI, 1,200-step horizon, ≈ 10 s decision interval
Serving/measurement GPU	NVIDIA RTX 2060, 6 GB (the deployment envelope)
Training GPU (3.8B)	UCL departmental NVIDIA RTX 4090, 24 GB
Training GPU (0.6B)	the RTX 2060 itself (13.5 h)
Host OS	Windows 10 (build 19045), which predates Windows ML EP autoregistration

Table 5: Version and hardware pins.

Family	Design	Primary output
Base-model factorial	8 catalogue SLMs $\times$ 4 formats $\times$ 6 topologies $\times$ up to 32 seeds	closed-loop delay vs baselines
Explanatory (why) study	two-way ANOVA, 2 models $\times$ 4 topologies $\times$ 3 seeds (config fixed)	F , η^2 , interaction
Teacher cross-audit	second frontier model re-labels 4 batches of 150 teacher-labelled states	label agreement
Offline distillation	7 phi rows + 4 qwen rows $\times$ 4 formats, frozen holdout	accuracy, JSON rate
Fine-tune closed-loop	ft vs stock at 0.6B, plus a 3.8B ft confirmation arm, $\times$ 3 topologies $\times$ 30 disjoint seeds	paired delay, authorship
Decision battery	cloud frontier vs local SLM on identical states	agreement, latency
Accident / crash-law suite	scripted collision scenarios $\rightarrow$ audit reconstruction	evidence packs, chain verification
Trust battery	honest / blatant / stealth misreporting	detection, delay impact
Emergency / EV suite	scripted EV arrival and spoofing attack on the arterial corridor, 3 modes $\times$ 30 paired seeds	clearance metrics

Table 6: Experiment inventory.

in `results/experiment_fixedtime_*.json`. Closed-loop figures recompute from both with the scripts in `src/`.

B.3 Failure forensics tables

The catastrophic-cell analysis of §6.7 is tabulated in `results/failure_forensics.md`: 14 of 270 cells at $\geq +20\%$ delay, their seed-matched comparators, and the shared-versus-arm-specific decomposition. The underlying data is in `results/failure_forensics.json`.

B.4 Adversarial trust battery

Per-topology, per-scenario cells for all three controllers (stock and fine-tuned, both scales) are in `results/experiment_trust_traffic_<model>.json`, with raw cells under `results/trust_traffic_raw/` keyed by model, topology, scenario, and seed.

B.5 Authorisation re-test

The four-model, two-arm fair re-test of §6.11 is recorded in `results/slm_auth_lee_<model>.json`, each containing class accuracy, grant accuracy, precision, recall, false-positive and false-negative rates, dangerous false grants,

parse failures, and P50/P95/P99 latency per arm.

B.6 Detail tables

Table 7 is the complete holdout table behind §6.3, including the LOTO and format-specialist controls; Table 8 the confidence intervals and rank tests behind §6.4; Table 9 the authorship, substitution and floor shares behind §6.6; Table 10 the per-cell latency summaries behind §6.12; and Table 11 the four emergency-vehicle quantities behind §6.10.

B.7 Backpressure guarantees and their assumptions

The four assumptions behind MaxPressure’s throughput-optimality [27] that §2.1 says fail in central London. First, the proof assumes unbounded queue storage: Gregoire et al. [7] prove that linear-pressure control of Varaiya’s form is not work-conserving once queue capacities are finite, construct explicit deadlocks, and repair it with a capacity-normalised pressure that is *not* part of standard MaxPressure; central London’s short links imply small capacities, so that failure condition is plausibly standing here, though I do not measure it. Second, the only stability result for a deployable (non-oracle) backpressure controller holds under heavy load

	myopic	sota	prediction	coordination
Qwen3-0.6B stock	45.6	55.1	63.3	69.3
Qwen3-0.6B generalist ft	87.9	97.7	97.6	99.0
Phi-4-mini stock	87.5	59.0	86.0	87.0
Phi-4-mini generalist ft	88.7	97.1	97.4	99.3
Phi-4-mini LOTO	88.3	96.8	97.4	99.1
Phi-4-mini sota-specialist	87.9	97.8	89.5	93.3
Phi-4-mini myopic-specialist	87.0	87.9	88.8	91.8
Phi-4-mini prediction-specialist	87.9	91.3	97.3	98.4
Phi-4-mini coordination-specialist	87.4	89.1	94.7	99.3

Table 7: Holdout accuracy (%) against frozen frontier-teacher labels, int4-as-served. All fine-tuned rows and the stock Phi row emit 100% strict JSON; stock Qwen3-0.6B reaches 93.4% on sota.

Arm	Topology	rel. (95% CI)	p_t	p_{Wilcoxon}
Qwen ft	Euston	+15.5 [−1.9, +32.9]	.079	.309
	Bloomsbury	+0.8 [−1.0, +2.6]	.355	.516
	Old Street	−11.6 [−17.2, −6.0]	.0002	3.2e-6
Qwen stock	Euston	+7.6 [−4.8, +20.0]	.221	.349
	Bloomsbury	+2.8 [+0.9, +4.6]	.005	.008
	Old Street	−9.7 [−13.3, −6.1]	5.1e-6	1.2e-6
Phi ft	Euston	+16.2 [−1.4, +33.9]	.070	.245
	Bloomsbury	+0.8 [−0.9, +2.5]	.365	.543
	Old Street	−9.7 [−15.4, −3.9]	.002	7.1e-5

Table 8: Closed-loop effects against the fixed-time floor, conventions as in Table 1, plus a Wilcoxon signed-rank test on the paired per-seed delays ($n=30$). Bold marks $p < .05$. The MaxPressure counterpart of every row is in Table 2.

alone: Gregoire et al. [6] require every queue to exceed saturation flow, concede that sub-saturation queues “can unstabilize the queuing network”, and call stability outside it “still a challenging problem”; measured diurnal demand sits below that threshold most of the day, and the candidate mechanism there, the pressure signal chasing arrival noise where a fixed cycle holds steady, I do not test. Third, every favourable empirical figure is a grid figure (roughly 90% of oracle backpressure on a uniform 21×21 grid, about 80% under heterogeneous parameters), and the authors caution that grid results cannot extend to arbitrary networks. Fourth, junctions with give-way movements, gap acceptance or gyratory circulation fall outside the model domain: service in these theorems is binary and phase-determined, so no guarantee applies.

B.8 Seed families outside the disjoint set

Three families of §5.3 do not use the training-disjoint seed set. The powered base-model factorial of §6.2 (seeds {1..30}) and the exploratory sweep that motivated it (three seeds per cell) involve no model fine-tuned in this project, so seed disjointness carries no leakage meaning for them. The trust battery of §6.9 does run fine-tuned controllers on training-state seeds (42 and 123, with 31 a disjoint evaluation seed) and is flagged there. Every other closed-loop *controller* number in §6 is a disjoint-seed number.

B.9 Longest-queue-first control

A third classical baseline tests whether the results of §6.1 are specific to *pressure*. Longest-queue-first serves the largest upstream queue with no downstream term, so unlike MaxPressure it will discharge into a blocked exit, though the two are routinely conflated. Over the same thirty seeds with shield, minimum green, clearance and anti-starvation floor identical, longest-queue-first is statistically indistinguishable from MaxPressure on all five topologies (−0.9% to +13.4%, every $p \geq .30$) and reproduces its pattern against the naive floor: worse on the corridor (+13.6%, $p=.027$), better on Old Street (−5.9%) and the calibrated grid (−8.3%). The pressure formulation is not detectably part of the classical weakness on a saturated arterial, though the intervals are wide (Euston calibrated [−12.7, +39.5]), so this is a non-rejection rather than a demonstrated equivalence; saturation, link storage and the assumed turning split all remain candidates.

B.10 Serving-artifact provenance

The nine-point gap of §6.2 is comparator-robust though the sign flip is not: against the fixed-time floor the catalogue and pipeline-compiled Qwen3-0.6B artifacts are +17.4% and +4.8%, both worse than the floor but 12.6 points apart, and the gap holds on two of the three topologies where I could compare them (Old Street −0.2%, $p=.93$).

Topology	Arm	SLM	shield	floor	diverg.
Euston peak-hour	MaxPressure	0.00	n/a	34.1	n/a
	qwen ft	0.64	0.00	35.7	0.12
	qwen stock	0.62	0.00	38.2	0.46
	phi ft	0.64	0.00	36.0	0.13
Bloomsbury grid	MaxPressure	0.00	n/a	28.4	n/a
	qwen ft	0.70	0.54	29.5	0.17
	qwen stock	0.62	0.00	37.6	0.46
	phi ft	0.70	0.00	29.6	0.17
Old Street	MaxPressure	0.00	n/a	43.1	n/a
	qwen ft	0.56	0.00	43.6	0.11
	qwen stock	0.39	0.00	61.2	0.60
	phi ft	0.57	0.00	43.2	0.12
Bloomsbury calib.	MaxPressure	0.00	n/a	24.3	n/a
	qwen ft	0.76	0.00	23.5	0.04
	qwen stock	0.59	0.00	41.4	0.56
	phi ft	0.77	0.00	23.3	0.04
Euston calib.	MaxPressure	0.00	n/a	36.3	n/a
	qwen ft	0.62	0.00	37.7	0.07
	qwen stock	0.59	0.00	40.6	0.37

Table 9: Decision composition for every closed-loop experiment ($n=30$ training-disjoint seeds per cell, sota format). *SLM* is the SLM-authored fraction of all logged decision ticks; *shield* and *floor* are percentages of those same ticks; *diverg.* is the fraction of *SLM-authored* decisions whose phase differed from MaxPressure’s. The *shield* column is the substitution path, the shield replacing an invalid or absent proposal with the MaxPressure phase; in the MaxPressure arm there is no SLM to substitute for, so the column does not apply and the classical policy authors every non-floor tick.

Arm	p50 (median)	p50 range	p99 (worst cell)	worst call
Qwen3-0.6B ft	1.75 s	1.39–2.57 s	7.58 s	13.79 s
Qwen3-0.6B stock	1.74 s	1.46–2.51 s	2.97 s	9.88 s
Phi-4-mini ft	0.94 s	0.92–1.20 s	1.46 s	1.49 s

Table 10: Decision latency over the three uncalibrated topologies (sota arms, GPU-served, $n=30$ disjoint seeds each): p50 is the median over cells of the per-cell median latency, the range spans the per-cell medians, p99 is the worst cell’s p99, and the worst call is the maximum single latency. The decision interval is 10 s.

B.11 Multiple comparisons

The rank-test results of §6.4 are not confirmatory: of roughly forty paired comparisons across the report, the two reaching $p < .05$ only under the rank test would not survive a Holm correction across that family, and I pre-registered no confirmatory comparison. The fine-tuned-versus-stock head-to-head is nonetheless the study’s primary contrast by design, the only cell in which a single variable is manipulated, and at $p=2.0 \times 10^{-21}$ it survives Holm where they do not.

B.12 Calibration decision audit

Is the demand calibration of §6.5 a principled correction or a search for a more favourable test bed? In favour: the guard metrics were recorded in every cell from the campaign’s first run, so their values were not produced by looking; the diagnosis is outcome-independent; and the 40% target matched Old Street’s completion rate, not any effect size.

Against: the decision to look followed a weak result, a data-dependent choice that retaining the superseded cells does not by itself redeem. Uniform application is the check I can offer: Euston peak-hour also fails the guard and is where SLM control looks worst (§6.2), so calibrating the corridor risked the conclusion rather than protecting it, and I calibrated it, though the probed fractions bracketed failure only on the grid. The rule, take the largest demand fraction that still clears, is stated in the text rather than encoded in the calibration script, which prints completion at each probed fraction and leaves the choice to the analyst.

B.13 Trajectory-level convergence detail

The identical cells of §6.7 are separate runs serving the same phase sequence, not duplicated records: wall-clock differs by up to $2.15\times$ and median latency by $2.23\times$, and in one both arms took 307 decisions with SLM/floor attribution differing

Quantity	Mean	95% CI
Ambulance time saved by priority (nopreempt – defended)	31.1 s	[25.4, 36.5]
Cost to the ambulance of the authorisation gate (defended – naive)	8.0 s	[3.7, 12.5]
Cross-traffic harm avoided under attack, mean	4.6 s	[3.3, 6.0]
Cross-traffic harm avoided under attack, worst case	16.1 s	[11.5, 20.4]

Table 11: Emergency-vehicle priority, $n=30$ seeds, 95% CIs. Positive values are seconds saved.

by exactly one tick, where floor and model chose the same phase. Of the 14 catastrophic cells, five are the stock arm alone, one the 3.8B student alone, and eight are four seed pairs on which both fine-tuned students fail together with per-seed relative delay agreeing to within 0.1–0.2 points; that pairing restates the convergence rather than explaining it, and the independent evidence is the decision-level agreement.

B.14 Completeness layer

Above the per-run chain sits the Certificate-Transparency-style completeness layer of §2.6. Over a twelve-entry batch, a salted Merkle root hides the entry contents while remaining provable, the salt is erasable off-chain, and an inclusion proof verifies while the same proof under a *forged* salt is rejected; under a two-witness cross-audit the witnesses agree, and when one is made to serve a different root the equivocation is detected and consistency fails, a witness refusing to overwrite its own history. The cross-auditor’s own honesty is an unverified assumption, on a par with the identity root, and the batch is not externally anchored: I withheld the public-log write rather than pollute a production transparency log.

B.15 Trust-coefficient battery detail

The property battery of §6.9 (9/9): a persistently honest junction climbs to 0.929 trust, a persistent liar collapses to 0.011 and loses the power to corroborate, the designed asymmetry holds in both directions, and a truthful-but-late report is vindicated retroactively by local sensing. In closed-loop traffic (honest, blatant and stealth misreporting injected into SUMO) the battery ran against fine-tuned 3.8B, stock 0.6B and fine-tuned 0.6B controllers. Trust gating was off, so the ledger altered no served phase: detection is scored by replaying each run’s recorded claims and local-sensing confirmations through the ledger, measuring what it would have caught rather than what it prevented, and the delay costs are the attack’s effect on an undefended controller. No stock-3.8B run exists, so the stock/fine-tuned contrast is available at 0.6B only. The battery runs three seeds (42, 123, 31) in the coordination format against the same-seed MaxPressure baseline; 42 and 123 are training-state seeds and 31 a disjoint evaluation seed (§5.3), making this the one family whose fine-tuned arms partly see states the students were distilled from. Every controller catches and collapses a blatant liar on Bloomsbury (62–70 catches, final trust 0.21–0.27) and grid4×4 (146–151 catches, trust ≤ 0.03 ,

corroboration stripped), while a calibrated stealth attacker evades essentially everywhere (≤ 2 catches, trust ≥ 0.92); on the daily-demand corridor, Euston peak-hour and Old Street the liar junction is never consulted, so the mechanism is inert there. Blatant lying adds +2.7 to +11.5 points over the same controller’s honest run on the synthetic grids even as the liar’s trust collapses; on Bloomsbury the damage stays within ~ 3 points (fine-tuned qwen -1.3% under attack against -1.1% honest), which the design predicts is the local-sensing override, but no override-ablated arm was run, so map structure is an equally live explanation. Stealth’s intermittent inflations cost -2.3 to $+2.9$ points, buying retained corroboration power at high trust to abuse later.

B.16 Emergency-vehicle priority detail

The three modes of §6.10 are *nopreempt* (no signal priority, as at an installation without emergency-vehicle detection), *naive* (preempts on any claim), and *defended* (preempts only on a claim with a valid authorising key and corroboration, §3.5). Table 11 gives the four quantities with 95% CIs over thirty seeds. Whether the exchange of 8.0 s of ambulance time for the protection the gate buys is worth making is a policy judgement.

B.17 Authorisation re-test detail

The legal-note pilot of §6.11 ran at 37.8 s per note, and its gold is policy-base-derived rather than human-graded, so that negative is a pilot proxy. The objection held that the negative could not stand until the SLM was re-tested with explicit in-prompt rules and a reason-then-classify structure, which the scaffolded arm supplies. The required output schema was {"class", "grant"}; the fine-tuned students’ training schema was {"phase": N}. The harness did not retain raw responses, so what ft-Qwen emitted instead of the schema cannot be shown.

B.18 Residual legal exposure

The audit scenarios use synthetic actors on real topology, so no real data subject exists, but a real deployment would face two residual exposures (§7.3): curating which rules apply is itself a judgement about relevance, so the evidence pack cannot be treated as a neutral readout, and I cite no authority for that proposition because I know of none; and the pack attracts only qualified privilege, which disclosure defeats.

B.19 Serving reliability and latency detail

The three reliability findings of §6.12 in full: one long offline evaluation died at 5,500 of 5,580 rows on a host `OSError(28)` (no space left on device) and had to be re-run; registering builder-compiled models required undocumented synthesis of the inference manifest; and the catalogue Qwen3-0.6B GPU artifact produced malformed output after extended service uptime, an unlogged operational observation rather than a measured finding, and the reason §5.4 takes the pipeline-compiled artifact as the control. Table 10 summarises decision latency. The 3.8B student is the fastest of my pipeline-compiled arms and the 0.6B students the slowest, a counter-intuitive ordering that is a property of the compiled artifacts rather than of parameter count and holds only within my own compiles: four vendor catalogue arms are faster still, including the catalogue build of the student's own weights at 0.68 s. One call in the 0.6B arms exceeded the 10 s decision interval; phi-4-mini-reasoning differs only in frequency, a lone outlier here against a p99 above the interval there.

C UK-law fault-grounding knowledge base

The reconstruction layer matches verified collision facts against a curated knowledge base of UK traffic law. The base is deliberately small, explicit, and citable: each rule carries a statute or case reference, a binding strength (`MUST` for statutory prohibitions, `common_law` for judicial doctrine), a fault weight, and the verified fact that triggers it. Table 12 lists the rules exercised by the seven evaluated scenarios.

One property of the encoding matters for the accountability claim: the emergency exemption is gated first on the mechanically verified crossing class, which a forged key cannot manufacture, and only then on corroboration or a valid emergency-class key, so a spoofed claim leaves the driver squarely within the red-light prohibition (§6.8.1). Legal causation is never assessed but marked `NOT_ASSESSED` for a human adjudicator.

D Deployment notes

Evidence, gaps and published cost figures behind §7.4.1. None of the prices or power figures below originates from this study's measurements.

Temporary and roadworks signals. Portable signals at a lane closure have no engineered plan, so a controller needing *no site configuration* is worth more there than one matching a tuned plan. By design, the phase model is derived from the live network at start-up (§3.1); measured, the leave-one-topology-out student matched the generalist across the mixed holdout (§6.3), evidence against topology memorisation in aggregate rather than parity on the withheld layout; and the anti-starvation floor enforces alternation deterministically except where it records a violation (§3.3), one requirement of shuttle working. Three gaps are clear: no lane-closure layout

was tested; the phase model is read once, so mid-deployment reconfiguration needs re-initialisation; and shuttle working's clearance-time constraint, the safety-critical one, belongs in the shield and is not implemented.

Emergency-vehicle priority across a corridor. Installations without emergency-vehicle detection give an approaching vehicle no priority, an omission §6.10 measures at 31.1 s per event. The substrate exists: junctions exchange Ed25519-signed reports over an MQTT adapter with a real broker between publisher and subscriber. Capability is not a deployment claim: the bus has run in simulation and over a local broker, not a field radio under loss and latency; the scenario is scripted, with the scope §6.10 declares, and corridor-scale progressive preemption is designed for but unmeasured.

Deployment economics. A market survey of published prices (September 2026)² places the compute node at well under 1% of site cost: a Jetson-class module lists at roughly £315–400 and runs this study's 0.6B student at a community-measured ~50 tokens/s within a 7–25 W envelope, against £80,000–130,000 for a signalised crossing. The power figure matters more than the price, since it answers the objection that my 6 GB desktop GPU misrepresents a cabinet power budget, on a published figure this study did not verify. TRL's SCOOT rollout averaged a 12.8% delay reduction, the bar any deployment would have to clear. Enclosure, certification, maintenance and integration dominate hardware at these ratios, so the whole-life case needs a costing study this report's data cannot supply.

²Vendor list prices, council committee papers and published trade estimates compiled in September 2026: NVIDIA's developer price list (July 2026), a Greater London Authority estimate for signalised crossings, TRL's SCOOT rollout evaluation, and INCE and UK MVNO tariffs. None originates from this study's measurements.

Rule	Authority	Effect
Driver crossed red	RTA 1988 s36	Criminal offence; driver fault high
Red prohibition	TSRGD 2016 Sch14	Must not proceed beyond stop line
Amber	TSRGD 2016 5(9)	Driver fault high where a safe stop was possible
EV exemption	TSRGD / conditional	Conditional on an emergency crossing class <i>plus</i> either corroboration or a valid emergency-class key; endangerment test still applies
Maintenance vehicle	no exemption	Works vehicle treated as an ordinary driver
Dark signal	Highway Code r.176	Duty to obey falls away; ordinary care governs
Authority misfeasance	<i>Bird v Pearce</i>	Positively-wrong indication: authority fault non-zero but low
Authority nonfeasance	<i>Gorringe v Calderdale</i>	Mere omission: authority fault defaults to zero
EV apportionment anchor	<i>Griffin v Mersey</i>	Default 60/40 against the non-emergency party
Green due regard	TSRGD 2016 Sch14 Pt1 5(14)	Green-light party must proceed with due regard; typically the minority share

Table 12: Fault-grounding rules exercised by the evaluated scenarios.